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United States Patent	12384532
Kind Code	B2
Date of Patent	August 12, 2025
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Vertical takeoff and landing (VTOL) aircraft systems and method

Abstract

An electrically-powered VTOL aircraft has a fuselage with a cabin flexibly connected to a powerplane assembly that includes a plurality of electrically-powered rotors and booms. The powerplane assembly can pitch and roll relative to and independently of the cabin, thereby generating efficient fore, aft and lateral thrust while the cabin attitude remains unchanged. This provides a stable passenger experience and enhanced performance and controllability with reduced cost and complexity. In some embodiments, the fuselage is vertically elongated and the powerplane assembly mounts above the fuselage such that a person may walk beneath the rotors completely erect. A VTOL docking station is also disclosed that is configured to allow the aircraft to land without the use of landing gear. The docking station also includes electrical components configured to automatically provide charging to the aircraft when the aircraft is docked with the docking station.

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Family ID:	89170263
Appl. No.:	18/461483
Filed:	September 05, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230406488 A1	Dec. 21, 2023

Related U.S. Application Data

continuation-in-part parent-doc US 17480890 20210921 US 11970264 child-doc US 18461483

Publication Classification

Int. Cl.: **B64C27/08** (20230101); **B64C27/52** (20060101); **B64D17/80** (20060101); **B64D27/34** (20240101); **B64D27/357** (20240101)

U.S. Cl.:

CPC **B64C27/08** (20130101); **B64C27/52** (20130101); **B64D17/80** (20130101); **B64D27/34** (20240101); **B64D27/357** (20240101);

Field of Classification Search

CPC: B64C (27/08); B64C (27/52); B64C (29/02); B64U (80/20); B64U (80/25); B64U (2101/61)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2016/0039541	12/2015	Beardsley	701/2	B60L 53/14
2023/0039483	12/2022	Strobel	N/A	B64U 10/20

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
212195854	12/2019	CN	B64C 1/30

OTHER PUBLICATIONS

Connecting structure & folding piece & folding unmanned machine (Year: 2020). cited by examiner

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application is a continuation in part of U.S. patent Ser. No. 17/480,890 filed on Sep. 21, 2021, which is incorporated herein by reference.

RELATED ART

(1) Aircraft design and configuration vary according to mission objectives and constraints. Electrically-powered vertical takeoff and landing (VTOL) aircraft have many benefits that make them attractive for use in short-range transport, especially in urban, suburban or parkland settings where suitable landing zones for conventional aircraft may be limited. For example, the ability to take-off vertically, hover, and in some cases to perform tight aerial maneuvers, enable such aircraft to operate safely around structures and in congested environments that are inaccessible to many

other types of aircraft. Furthermore, electric aircraft can be designed to operate more quietly than other aircraft, allowing electric aircraft to operate in populated areas without violating applicable noise regulations.

(2) A wide range of electrically-powered VTOL aircraft designs have been introduced in recent years, primarily focused on passenger or small cargo transport in environments where ground-based traffic congestion makes such aircraft an attractive alternative. While designs vary greatly according to mission objectives, most fall into one of three broad categories: Multicopter (aka Multicopter) solutions are simple and cost-effective, but have limited range and generate horizontal thrust by tilting the airframe, which may be uncomfortable for some passengers. Lift+Cruise solutions increase range and payload by adding wings and a separate forward propulsion system, but in so doing may lose maneuverability and add weight, cost and complexity. Tilt Rotor solutions (aka Vectored Thrust) also add wings to increase range and payload, but use tilting rotors or wing/rotor assemblies to generate both vertical and horizontal thrust from individual propulsion units, again at the price of complexity, cost, and reliability concerns. In summary, it is challenging to design electrically-powered VTOL aircraft that combine performance, safety, and a stable and reassuring passenger experience in a cost-effective, maneuverable package suitable for use in congested environments. Thus, alternative or improved designs of eVTOL aircraft are generally desirable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure can be better understood with reference to the following drawings. The elements of the drawings are not necessarily to scale relative to each other, emphasis instead being placed upon clearly illustrating the principles of the disclosure. Furthermore, like reference numerals designate corresponding parts throughout the several views.

(2) FIG. 1A depicts a perspective view of an embodiment of a vertical takeoff and landing (VTOL) aircraft.

(3) FIG. 1B depicts a left side view of the VTOL aircraft depicted by FIG. 1A.

(4) FIG. 1C depicts a front view of the VTOL aircraft depicted by FIG. 1A.

(5) FIG. 1D depicts a right side view of the VTOL aircraft depicted by FIG. 1A.

(6) FIG. 2 depicts a side view of the VTOL aircraft of FIG. 1 with portions of the fuselage removed to expose components within the fuselage.

(7) FIG. 3 depicts a top view of the VTOL aircraft depicted by FIG. 1A.

(8) FIG. 4 is a block diagram illustrating an embodiment of a flight control system for controlling VTOL aircraft, such as is depicted by FIG. 1A.

(9) FIG. 5 depicts a side view of the VTOL aircraft of FIG. 1A with a powerplane assembly of the aircraft positioned with aft pitch.

(10) FIG. 6 depicts a side view of the VTOL aircraft of FIG. 1A with a powerplane assembly of the aircraft positioned with forward pitch.

(11) FIG. 7 depicts a front view of the VTOL aircraft of FIG. 1A with a powerplane assembly of the aircraft tilted about the aircraft's pitch axis and roll axis.

(12) FIG. 8A depicts a perspective view of an embodiment of a power module.

(13) FIG. 8B depicts a perspective view of the power module depicted by FIG. 8A with access panels removed.

(14) FIG. 8C depicts a perspective view of the power module depicted by FIG. 8A, showing internal components including batteries and motor controllers.

(15) FIG. 9 depicts a perspective view of an embodiment of a ballistic recovery system.

(16) FIG. 10 depicts a harness for a ballistic recovery system, such as is depicted by FIG. 9.

(17) FIG. 11A depicts the VTOL aircraft of FIG. 1A as parachutes of a ballistic recovery system are being deployed.

(18) FIG. 11B depicts the VTOL aircraft of FIG. 11A after the depicted parachutes have been further deployed.

(19) FIG. 11C depicts the VTOL aircraft of FIG. 11B once the parachutes have fully deployed.

(20) FIG. 12 depicts a perspective view of a base of the fuselage of the VTOL aircraft depicted by FIG. 1A.

(21) FIG. 13A depicts a side view of landing gear coupled to the base of the fuselage depicted by FIG. 12.

(22) FIG. 13B depicts an elastic coupling of a suspension system for the landing gear depicted by FIG. 13A.

(23) FIG. 13C shows the landing gear of FIG. 13A after foot of the landing gear has been moved relative to the fuselage 25.

(24) FIG. 13D depicts the elastic coupling of FIG. 13B after the foot of the landing gear has been moved, as shown by FIG. 13C.

(25) FIG. 14A depicts a bottom perspective view of a landing gear leg with self-catering roller, such as is depicted by FIG. 1A.

(26) FIG. 14B depicts a cross-sectional view of the landing gear foot depicted by FIG. 14A.

(27) FIG. 15A depicts a perspective view of a pivotal coupling that connects a powerplane assembly to a power module of VTOL aircraft, such as is depicted by FIG. 1A.

(28) FIG. 15B depicts an exploded view of the pivotal coupling depicted by FIG. 15A.

(29) FIG. 16 depicts a perspective view of the pivotal coupling depicted by FIG. 15A.

(30) FIG. 17 depicts a side view of the pivotal coupling depicted by FIG. 16.

(31) FIG. 18 depicts a cross-sectional view of the pivotal coupling depicted by FIG. 16 taken along the cut lines shown by FIG. 17.

(32) FIG. 19 depicts a cross-section view of the pivotal coupling depicted by FIG. 16 taken along the cut lines shown by FIG. 18.

(33) FIG. 20 depicts a perspective view of a support structure for connecting booms of a powerplane assembly to the pivotal coupling depicted by FIG. 16.

(34) FIG. 21 depicts a perspective view of a boom connected to a motor and rotor assembly for VTOL aircraft, such as is depicted by FIG. 1A.

(35) FIG. 22 depicts a perspective view of an embodiment of a VTOL aircraft.

(36) FIG. 23 depicts a perspective view of an embodiment of a VTOL docking station.

(37) FIG. 24 is a block diagram illustrating an embodiment of a flight control system for controlling VTOL aircraft, such as depicted in FIG. 22.

(38) FIG. 25 depicts a perspective view of a VTOL aircraft docked at a VTOL docking station, such as depicted by FIGS. 22-23.

(39) FIG. 26 depicts an exploded view of the VTOL docking station, such as depicted by FIG. 23.

(40) FIG. 27A depicts a powerplane assembly with electrical contact strips located on a lower surface of each boom.

(41) FIG. 27B depicts an upper surface of the VTOL docking station including one or more electric contact strips for forming an electrical connection to a docked VTOL aircraft, such as depicted by FIG. 22.

(42) FIG. 27C is a diagram representing how electrical contact strips of a powerplane assembly can contact the one or more electric contact strips of a VTOL docking station with the VTOL aircraft in various docking positions.

(43) FIG. 28A depicts a boom of a VTOL aircraft, such as depicted by FIG. 22.

(44) FIG. 28B depicts a profile perspective of a boom, such as depicted by FIG. 28.

(45) FIG. 28C depicts a profile perspective of a boom, such as depicted by FIG. 21.

(46) FIG. 29 depicts a perspective view of a boom mount exoskeleton that connects a powerplane

assembly to a power module of the VTOL aircraft, such as depicted by FIG. 22.

(47) FIG. 30 depicts a perspective view of a boom mount endoskeleton that connects a powerplane assembly to a power module of the VTOL aircraft, such as depicted by FIG. 22.

(48) FIG. 31 depicts an embodiment of a pivotal coupling, such as depicted by FIGS. 29-30.

(49) FIG. 32 illustrates an exemplary use of a VTOL aircraft system, according to aspects of the present disclosure.

(50) FIG. 33 illustrates another exemplary use of a VTOL aircraft system, according to aspects of the present disclosure.

(51) FIG. 34 illustrates yet another exemplary use of a VTOL aircraft system, according to aspects of the present disclosure.

(52) FIGS. 35A-D illustrate the effect of varying a center of mass of a VTOL aircraft, according to aspects of the present disclosure. More specifically, FIG. 35A depicts an exemplary VTOL aircraft having a low center of mass in a stationary position, FIG. 35B depicts the aircraft of FIG. 35A beginning lateral flight, FIG. 35C depicts the aircraft of FIG. 35A in lateral flight, and FIG. 35D shows a VTOL aircraft having a higher center of mass in lateral flight.

DETAILED DESCRIPTION

(53) The present disclosure generally relates to VTOL aircraft systems and methods, and various embodiments are described herein. In some embodiments of the present disclosure, an electrically-powered VTOL aircraft has a fuselage that is vertically elongated with a plurality of electrically-powered rotors mounted above the fuselage. The rotors are mounted sufficiently high above the ground such that a person may safely walk completely erect beneath the rotors without being struck by any of the rotors. Each rotor motor is mounted to one end of a boom, with the other end of the boom joined to a central connector, and together the rotors, motors, booms and central connector form a powerplane assembly. The powerplane assembly is joined to the fuselage by a pivotal coupling, which enables the powerplane assembly to pitch and roll independently of the fuselage.

(54) The vertically elongated fuselage with elevated powerplane assembly offers a wide range of benefits including (but not limited to) fast, easy entry and exit; enhanced passenger and pedestrian safety during ground operations, maintenance, boarding and flight; minimal ground footprint independent of rotor or powerplane diameter; reduced rotor wash interaction with the ground, buildings or structures; reduced susceptibility to rotor damage due to ground handling or recirculated debris; a minimized planform silhouette which makes the flying vehicle less obtrusive to ground observers; and a 360° view for passengers.

(55) In some embodiments disclosed herein, a multicopter VTOL aircraft comprises a fuselage that is pivotally coupled to a separate powerplane assembly having a plurality of electrically-powered rotors joined by booms and a boom mount so as to form a single structural unit. In some embodiments, the pivotal coupling comprises a biaxial joint which enables the fuselage to hang freely from the powerplane assembly in flight, so that the powerplane assembly is free to pitch and roll independently of the fuselage, but remains indexed to the fuselage in yaw. By this means the powerplane assembly is able to tilt and generate improved thrust efficiency in any compass direction of the x/y plane without the cost or complexity of an actuating mechanism, while also offering an improved passenger experience by maintaining a substantially constant fuselage attitude with respect to ground. In this regard, free movement of the powerplane assembly about the pitch and roll axis not only helps to reduce tilting of the fuselage, but it also allows the aircraft to move directly from one point to another along any combination of x, y, and z vectors without changing the cabin attitude or heading, and it allows the aircraft to precisely hold position in 3D space even in a gusty wind environment. Thus, cabin tilt and heading changes can be reduced during maneuvering, providing a more relaxing flight for the passenger, and the aircraft can safely and precisely follow complex, multi-segment 3D flight paths in obstacle-rich urban environments where dynamic wind conditions are common. Some additional benefits of this embodiment include isolation of the fuselage from powerplane-induced vibration; simplified structural loading of the

fuselage by the elimination of rotor-induced bending loads; and quickened control system responsiveness resulting from the reduced mass and moments of inertia of the powerplane assembly compared with a complete, non-pivoting airframe. In addition, due to the separation of the powerplane assembly from the fuselage by the pivotal coupling, the vertical orientation of the fuselage is subject to tensile and compressive loads, with little bending input. This load regime simplifies weight optimization and is ideal for carbon fiber structures.

(56) Thus, VTOL aircraft systems in some embodiments combine a simple, robust and reliable method of enhancing multirotor performance with a stable, reassuring passenger experience during take-off, transition, and cruise, thereby realizing a simple, efficient, short-range, electrically-powered VTOL aircraft that provides enhanced performance and a comfortable passenger experience with reduced manufacturing, operating, maintenance and logistics costs.

(57) FIGS. 1A-1D depict an exemplary embodiment of a VTOL aircraft **20** that may be used to transport passengers or cargo. The aircraft **20** has a fuselage **25** that is vertically elongated (in the z-direction). For illustrative purposes, it will be assumed herein that the x-direction shown in the drawings is parallel to the roll axis, the z-direction shown in the drawings is parallel to the yaw axis, and the pitch axis (parallel to a y-direction) is perpendicular to both the x-direction and the z-direction shown in the drawings.

(58) As an example, the fuselage **25** may be vertically elongated such that it has an aspect ratio ($\frac{h_{sub.f}}{w_{sub.f}}$), referred to hereafter as “fuselage aspect ratio,” of about 2.0 or greater, where “ $h_{sub.f}$ ” is the maximum height of the fuselage **25** in the z-direction and “ $w_{sub.f}$ ” is the maximum width of the fuselage **25** in the x-direction. However, a greater fuselage aspect ratio may help to increase the height of the powerplane assembly **33**, thereby realizing various benefits, as will be described in more detail below. To enhance at least some of these benefits, it may be desirable for the fuselage aspect ratio to be at least 3.0 or even 4.0 or more. For example, a greater fuselage aspect ratio may help to minimize or eliminate downwash ground effect interactions, thus simplifying control system workload and enhancing passenger comfort. Having a greater rotor height not only provides some performance benefits, as described in more detail below, but also may help to provide passengers with a greater sense of safety. It may also reduce the rotor wash felt by a passenger or other user while walking beneath the rotors **52**.

(59) The vertical orientation of the fuselage **25** also helps to optimize the shape of the fuselage **25** for streamwise aerodynamic efficiency and reduced VTOL drag. Some additional benefits include: provide more room to accommodate one or more standing passengers and provide easy entry and exit to the cabin for passengers; reduce the ground footprint, helping to facilitate maneuvers in confined or congested areas; and provide greater vertical cabin space to accommodate long, unwieldy cargo. In other embodiments, aspect ratios below 2.0 are possible, and it is unnecessary for the fuselage **25** to be vertically elongated. Indeed, although the vertical orientation of the fuselage **25** has many advantages, as described herein, it should be emphasized that such vertical orientation is unnecessary in all embodiments, and any shape of the fuselage **25** is possible in other embodiments.

(60) The fuselage **25** has one or more modules, such as a cabin **31**, a powerpack assembly **32**, a powerplane section **35**, and a nose **34** (which is tapered in the shape a cone, though other shapes are possible). The fuselage **25** is coupled to a powerplane assembly **33**, which includes booms **51** and rotors **52**, used to provide lift and controllability as will be described in more detail below. In some embodiments, the walls of the fuselage **25** are composed of a carbon fiber material helping to reduce weight. As shown by FIG. 1A, the aircraft **20** has landing gear comprising a plurality of legs **36** that extend from the bottom of the fuselage **25** and make contact with the ground when the aircraft **20** lands. In one embodiment, the aircraft **20** has four such legs **36**, but other numbers of legs **36** less than or greater than four are possible in other embodiments. In the embodiment depicted by FIG. 1A, the aircraft **20** is dimensioned such that, when it is on the ground prior to takeoff, the bottom of the cabin **31** is about 1.2 feet above the ground, the bottom of the powerpack

assembly **32** is about 9.1 feet above the ground, the bottom of the powerplane section **35** is about 16.6 feet above the ground, and the top of the nose **34** is about 18.1 feet above the ground such that the overall height of the fuselage **25** is about 18.1 feet above the ground.

(61) In some embodiments, the aircraft **20** is designed to have a cruise speed of up to about 60 miles per hour with a weight of about 900 pounds and a targeted service range of about 10 miles (with reserve). In addition, the aircraft **20** is designed to produce noise during takeoff of less than 60 decibels (dB) from a distance of 50 feet. In other embodiments, other design and performance characteristics are possible.

(62) The cabin **31** forms a compartment in which cargo, passengers, and/or other types of payloads may be situated for transport, and will sometimes be referred to hereafter as “passenger cabin.” In the embodiment depicted by FIG. 1A, the cabin **31** is designed to accommodate a single passenger who stands on a floor **38** (FIG. 2) of the cabin **31** during operation. Also, the cabin **31** has a door **41** (FIG. 1D) through which the passenger may enter the cabin **31**, and the cabin **31** has a plurality of windows **43** through which the passenger may see outside the cabin **31**. In other embodiments, the cabin **31** may be designed to accommodate a greater number of passengers and/or other types of payloads, such as cargo (e.g., one or more packages to be delivered to one or more locations), and use of windows **43** is unnecessary. In addition, the cabin **31** may include one or more seats for permitting one or more passengers to sit as may be desired during transport.

(63) The vertical orientation of the fuselage **25** and, in particular, the cabin **31** allows a passenger to stand or maintain an upright posture, which may be preferred or more enjoyable to the passenger, enables rapid entry and exit, and is easy to clean between flights. Further, as will be described in more detail below, the aircraft **20** may be designed such that the cabin **31** experiences very little or no movement about the pitch axis and roll axis in response to differential thrusting of the rotors **52**, thereby reducing or eliminating perturbations or movements experienced by the passenger during flight. The vertical orientation is also conducive to a simple but effective ballistic recovery system (BRS) design, which will be described in more detail below. The BRS may be configured to deploy one or more parachutes in such a way that the orientation of the cabin **31** during parachute recovery remains substantially similar to its orientation during normal flight, thereby helping to reassure the passenger during an emergency event. Also, the BRS may be located above the powerplane assembly **33**, helping to keep the parachute lines clear of the powerplane rotors **52**. In addition, the parachute opening shock loads are fed into the airframe in pure tension, simplifying airframe structural design.

(64) In some embodiments, the shape of the vertically elongated fuselage **25** is streamlined so as to reduce aerodynamic drag from air being blown over the fuselage **25** by the rotors **52**. In the embodiment depicted by FIG. 1A, the fuselage **25** has a generally circular cross-section along its longitudinal axis (which is parallel to the z-direction in the exemplary embodiment of FIG. 1A), and the fuselage **25** may be tapered. As an example, and as shown by FIG. 1A, the fuselage's diameter or width may increase downward in the z-direction from the top of the fuselage **25** in order to provide greater area for the powerpack assembly **32** and the cabin **31**. However, the diameter of the lower portion of the cabin **31** slightly decreases toward the base of the cabin **31**. In some embodiments, the maximum diameter or width of the cabin **31** is about 38 inches, but other maximum cabin widths are possible. In other embodiments, the fuselage **25** may have other types of profiles and contours, and as noted above, the pivotally-coupled powerplane assembly described herein is compatible with a wide range of vertically or horizontally elongated fuselage configurations.

(65) The shape and dimensions of the cabin **31** are designed such that a single passenger may comfortably stand fully erect in the cabin **31** during flight. In some embodiments, the interior height of the cabin **31** from floor to ceiling is about 7.9 feet, though other heights are possible in other embodiments. Indeed, heights above about 6.5 to 7.0 feet will accommodate the heights of most passengers in order to allow them to stand fully erect in the cabin **31**. In some embodiments,

the cabin **31** may have one or more holding apparatuses, such as a handrail (not shown) or hand strap, mounted on an interior wall of the cabin **31**, and a passenger may hold onto such holding apparatus during flight in order to help maintain balance. To provide more room for standing, the cabin **31** may be designed without a passenger seat, but it is possible for the cabin **31** to have one or more seats in which one or more passengers may sit during flight. Yet other designs and configurations of the cabin **31** are possible in other embodiments.

(66) The powerplane assembly **33** comprises a plurality of booms **51** extending from a central mount and a plurality of rotors **52** respectively mounted on the booms **51**. In the embodiment depicted by FIGS. **1A** and **3**, the powerplane assembly **33** has four booms **51** and four rotors **52** with five blades each, but the powerplane assembly **33** may have other numbers of booms **51**, rotors **52**, or rotor blades in other embodiments. In the embodiment depicted by FIGS. **1A** and **3**, the rotors **52** provide sufficient lift for the aircraft **20** without the need of other lift-generating apparatuses, such as wings. However, in other embodiments, the aircraft **20** may have winglets or other lift-generating apparatuses, which may help to increase the range of the aircraft **20**.

(67) The rotors **52** are driven by motors **66** (FIG. **2**) that cause the rotors **52** to generate aerodynamic thrust to be used for generating lift and controlling powerplane pitch, roll, and yaw. In this regard, FIG. **4** shows an exemplary flight control system **60** of the aircraft **20** that can be used to control movement of the aircraft **20**. In the embodiment depicted by FIG. **4**, the system **60** comprises a flight controller **63** that is coupled to a plurality of motor controllers **65**, which are configured to control motors **66** for driving the rotors **52**. The flight controller **63** is also coupled to a plurality of sensors **69** that are configured to sense flight characteristics of the aircraft **20**, such as horizontal airspeed, vertical airspeed, heading, powerplane attitude, and other flight characteristics that are typically used to control flight of conventional aircraft. The sensors **69** may also sense other parameters useful in controlling the aircraft **20**, such as external objects for sensing and avoiding collision threats, temperature, etc. The flight controller **63** is further coupled to a power source **75**, which may include one or more batteries for powering the electrical motors and components of the aircraft **20**, and a ballistic recovery system (BRS) **72**.

(68) The flight controller **63** is configured to control the general operation of the aircraft, including flight, as will be described in more detail below. As an example, the flight controller **63** may be configured to provide control inputs to the motor controllers **65** for controlling the spinning speeds of the blades of the rotors **52** such that they generate thrust as appropriate for controlling the movement of the aircraft **20** in a desired manner. That is, the flight controller **63** instructs each motor controller **65** to operate its respective rotor **52** at a desired speed so that appropriate thrusts are generated for controlling the flight of the aircraft **20**. As an example, the rotors **52** may be controlled to generate sufficient lift to counteract the weight of the aircraft **20** such that the aircraft **20** is flown to a desired altitude above ground. Further, differential thrusting of the rotors **52** may be used to control movement of the powerplane assembly **33** about the roll and pitch axes, as is known in the art for conventional multirotor VTOL aircraft, and also to provide horizontal movement of the aircraft **20** as may be desired. Further, differential torque from the rotors **52** may be used control movement of the powerplane assembly **33** about the yaw axis, as is known in the art for conventional multirotor VTOL aircraft.

(69) In addition, in some embodiments, as will be described in more detail below, the powerplane assembly **33** may be coupled to the fuselage **25** via a pivotal coupling **152** (FIG. **15A**) that permits the powerplane assembly **33** to pivot about at least two axes (e.g., pitch and roll axes) relative to the fuselage **25**. In such an embodiment, the differential thrusting used to rotate the powerplane assembly **33** about the roll and pitch axes does not cause corresponding movement of the fuselage **25**. That is, in some embodiments for which the pivotal coupling **152** couples the powerplane assembly **33** to the fuselage **25**, the powerplane assembly **33** rolls and pitches independently of the fuselage **25**. However, in some embodiments the powerplane assembly **33** and the fuselage **25** may be locked about the yaw axis such that the fuselage **25** and the powerplane assembly **33** rotate

about the yaw axis together. An exemplary pivotal coupling **152** for achieving this effect will be described in more detail below with reference to FIGS. **15A** and **16-19**. However, it should be noted that use of such a pivotal coupling **152** is unnecessary, and other types of couplings are possible in other embodiments. As an example, if desired, the fuselage **25** may be designed such that the cabin **31** rotates with the powerplane assembly **33** about roll and/or pitch.

(70) FIG. **6** shows a side view of an embodiment for which the powerplane assembly **33** is coupled to the fuselage **25** through a pivotal coupling **152** (not shown in FIG. **6**) that permits the power plane assembly **33** to move independently about roll and pitch relative to the fuselage **25**. In FIG. **6**, the powerplane assembly **33** has forward pitch resulting from the rotors **52** at the back of the aircraft **20** being controlled to generate more thrust than the rotors **52** at the front of the aircraft **20**. In such a configuration, a horizontal component of the thrust generated by the rotors **52** pushes the aircraft **20** forward. FIG. **5** shows a side view of an embodiment having aft pitch resulting from the rotors **52** at the front of the aircraft **20** being controlled to generate more thrust than the rotors **52** at the back of the aircraft **20**. In such a configuration, a horizontal component of the thrust generated by the rotors **52** pushes the aircraft **20** backward.

(71) Similarly, the powerplane assembly **33** may be controlled to tilt about the roll axis in a desired manner by controlling the rotors **52** on one side of the aircraft **20** to generate more thrust than the rotors **52** on the other side of the aircraft **20**. In such a configuration, a horizontal component of the thrust generated by the rotors **52** pushes the aircraft **20** sideways.

(72) Moreover, through differential thrusting as described above, the thrusts generated by the rotors **52** may be controlled such that the powerplane assembly **33** may be simultaneously tilted about the pitch axis and roll axis as desired. For example, FIG. **7** shows the powerplane assembly **33** having about 20 degrees forward pitch and about 10 degrees of roll, thus generating a combined diagonal x/y thrust resultant. Notably, the powerplane assembly **33** is coupled to the fuselage **25** in the Z axis, so that motor torque differential may be used to control aircraft yaw as is known in the art.

(73) The flight controller **63** may be implemented in hardware or a combination of hardware with software and/or firmware. As an example, the flight controller **63** may be implemented with hardware gate logic, such as a field programmable gate array (FPGA), or with application-specific integrated circuits (ASICs). In some embodiments, the flight controller **63** comprises at least one processor, such as a digital signal processor (DSP) or a central processor unit (CPU), that is configured to execute software and/or firmware for performing the functions described herein for the flight controller **63**. In other embodiments, other types of controllers **63** may be used.

(74) In some embodiments, the motors **66** are electrical and draw electrical power from the power source **75** during operation. The motors **66** may operate under the control of one or more motor controllers **65**, which control the motors **66** in order to control the spinning speeds of the blades of the rotors **52** as instructed by the flight controller **63**. In some embodiments, the aircraft **20** is autonomous, and the flight controller **63** controls flight of the vehicle **20**, including both navigation and aviation, without the use of or inputs from a pilot. However, in other embodiments, a pilot onboard the aircraft **20** may provide control inputs for controlling flight. Such inputs may be received by the flight controller **63**, which interprets such inputs to provide appropriate control signals to the motor controllers **65**. If desired, such inputs may be received from a remote pilot via wireless communication. Yet other techniques and configurations for controlling the flight of the aircraft **20** are possible in other embodiments. It is also possible to drive the rotors **52** with other types of motors.

(75) The powerpack assembly **32** houses the power source **75** (FIG. **4**) and motor controllers **65**, as shown by FIGS. **8A** and **8B**. In one embodiment, the power source **75** comprises a plurality of battery packs **80** that are positioned just below the motor controllers **65**, as shown by FIGS. **8B** and **8C**. In other embodiments, other types of power sources **75** may be used, and the relative positions of the battery packs **80** and the motor controllers **65** may be different. In the embodiment shown by FIGS. **2**, **8A**, and **8B**, the powerpack assembly **32** is located below the powerplane assembly **33**,

including the motors **66** and rotors **52**. By locating the battery packs **80** and motor controllers **65** just below the powerplane assembly **33**, the lengths of electrical wiring between the battery packs **80**, motor controllers **65**, and rotor motors **66** can be kept relatively short, thereby helping to minimize the electrical resistance and weight of such wiring. Additionally, positioning the power source **75** near the powerplane pivotal coupling **152** helps to maintain a vertical cabin orientation during lateral acceleration, as will be further described with respect to FIG. **35**. In some embodiments, the powerpack assembly **32**, which comprises the battery packs **80**, the motor controllers **65**, and the flight controller **63**, can be removed from the fuselage **25** and replaced with a new powerpack assembly **32**.

(76) In one embodiment, the powerpack assembly **32** comprises 812 battery cells providing a total power of approximately 71 kilowatt hours (kWh), although other numbers of cells and amounts of available power are possible. These cells are arranged in several interconnected battery packs **80** where each pack **80** comprises a plurality of cells. In the embodiment shown by FIG. **8C**, the battery packs **80** are arranged into two vertically-stacked towers **79**. The battery packs **80** may be stacked in other arrangements, and other numbers of battery pack towers **79** may be used in other embodiments.

(77) In some embodiments, each battery pack tower **79** is electrically connected to each motor **66** for driving the rotors **52** so that each motor **66** may receive power from either or both of the towers **79**. In addition, each battery pack tower **79** is able to generate sufficient power to drive simultaneously all of the motors **66** without power from the other tower **79**, if desired. Thus, in the event of an operational problem with one of the towers **79**, the other tower **79** may provide sufficient power for all of the motors **66** for at least enough time for the aircraft **20** to continue to the destination or perform an emergency landing. However, in other embodiments, other types and arrangements of batteries may be used.

(78) As shown by FIGS. **1C** and **15B**, an intake duct **82** is located in the powerpack assembly **32** of the fuselage **25** to receive cooling air from the rotors **52**. The air travels from this intake duct **82** through the powerpack assembly **32** and exits the powerpack assembly **32** via an outlet duct **85** (FIGS. **1B** and **1D**) that is located on an opposite side of the fuselage **25** relative to the intake duct **82**. The outlet duct **85** may be fed by a cooling fan, creating negative pressure within the powerpack assembly **32** to aid the intake flow. As the air is passing through the powerpack assembly **32**, it absorbs heat from the battery packs **80** and other electronics, and then is expelled through the outlet duct **85**, thereby ensuring the electronic equipment is maintained at its designed operating temperature.

(79) In some embodiments, the powerpack assembly **32** can be cooled via liquid cooling. For example, the fuselage **25** can be equipped with a heat exchanger (not shown) filled with a cooling liquid that circulates the cooling liquid through liquid channels disposed within the powerpack assembly **32**. In such embodiments, air traveling from the intake duct **82** can pass through the heat exchanger and out of the fuselage **25** via outlet duct **85**. As the cooling liquid circulates through powerpack assembly **32**, heat is drawn into the cooling liquid and is subsequently dispersed into the surrounding environment as the cooling liquid passes through the heat exchanger, in which the cooling liquid is cooled by the air traveling between intake duct **82** and outlet duct **85**. In some embodiments, the heat exchanger may additionally be equipped with a fan in order to increase the rate of heat exchange between the cooling liquid within the heat exchanger and the air of the surrounding environment, for example during charging of battery pack **80**.

(80) In the event of a fire in the powerpack assembly **32**, the vertical orientation of the fuselage **25** ensures that the passenger cabin **31** is well positioned below the smoke, heat or flame that may rise from such a fire. Additionally, the passenger cabin **31** is separated from the battery packs **80** by two bulkheads, each of which may be designed according to applicable aviation fire-resistance regulations, in order to ensure that the aircraft **20** reaches the ground and the passenger safely exits before the fire is able to penetrate the cabin **31**. For example, from an altitude of 500 feet above

ground level (AGL) with a BRS descent rate of 10 fps, the fire-resistant bulkhead system would need to remain intact for at least 50 seconds, well within the capabilities of modern fire-resistant materials.

(81) As shown by FIGS. 9 and 10, in some embodiments, the aircraft 20 may have a ballistic recovery system (BRS) 72 that can deploy one or more parachutes 135 (FIGS. 11A-11C) in the event of a failure of one or more components (e.g., rotors 52) or other operational problem. The BRS 72 has a hollow container 102, referred to hereafter as “chute can,” for holding the parachutes 135. Mounted on the chute can 102 are one or more rocket holders 105 (FIG. 10) for holding rockets (not shown in FIGS. 9 and 10) that are deployed in the event of an emergency event, such as failure of one or more rotors 52 or detection of a fire. When an emergency event is detected by the flight controller 63, it transmits a control signal for causing the rockets in the rocket holders 105 to be deployed. In response, the rockets are activated so that they fire out of the holders 105 upward (generally in the z-direction) either breaking through the nose 34 or forcing the nose 34 off of the aircraft 20 and pulling the parachute or parachutes 135 out of the chute can 102. The parachutes 135 then open to provide significant drag as the aircraft 20 falls to the ground. FIGS. 11A-11C depict a progression of three parachutes 135 being deployed. In other embodiments, other numbers of parachutes 135 may be used.

(82) As shown by FIGS. 9 and 11A-11C, each parachute 135 is coupled by one or more parachute lines 137 to the cabin 31 via a cabin bridle 110, as will be described in more detail below. Since the parachute lines 137 are coupled to the top of the cabin 31 as shown, deployment of the parachutes 135 should not materially change the attitude of the cabin 31, helping to make parachute deployment relatively smooth for the passenger.

(83) In some embodiments, the parachutes 135 provide sufficient drag to limit the descent rate of the aircraft 20 to about 10 feet per second (fps). Thus, at low altitudes of about 500 feet above ground or less, the aircraft 20 should reach the ground in under 50 seconds after parachute deployment. In other embodiments, parachutes 135 providing other descent rates and other types of recovery systems for recovering from an emergency condition are possible.

(84) As noted above and referring to FIG. 10, parachute lines 137 may connect to the cabin 31 through a cabin bridle 110. In this regard, the cabin bridle 110 has a plurality of cables made of high-strength steel wire rope that pass around the booms 51 and thru the powerpack assembly 32 (FIG. 1A) and connect to the passenger cabin 31, thus establishing a direct connection between each parachute 135 and the passenger cabin 31. In the embodiment depicted by FIG. 10, the bridle 110 comprises a plurality of cables 114 that connect the parachute lines 137 to a cabling loop 117 that connects to cables 122, which pass thru the powerpack assembly 32 and connect directly to the passenger cabin 31.

(85) In some embodiments, the parachute lines 137 may couple to the bridle 110 through a coupling assembly having one or more couplers 143 that may be mounted on, attached to, or pass through the chute can 102 and that connect the cables 114 to the parachute lines 137. The material selected for the cabling loop 117 and the cables 114, 122 can be capable of withstanding heat from a fire over a sufficient period of time to allow the aircraft 20 to reach the ground. As noted above, for altitudes of about 500 feet or less, it may take less than 50 seconds for the aircraft 20 to reach the ground after detection of an emergency condition and deployment of the parachute.

(86) In some embodiments, each cable 114, 122 may be designed to support the expected weight of the aircraft 20 such that the BRS 72 remains operational for keeping the cabin 31 connected to the parachutes 135 as long as at least one cable 122 and at least one cable 114 remain connected. Thus, if a fire or other event severs other cables 114 and 122, the cabin 31 can still reach the ground at a safe speed. In other embodiments, other types of harnesses 72 and other harness configurations may be used. In addition, other types of recovery systems are possible, and the use of a BRS 72 or other type of recovery system is unnecessary in all embodiments.

(87) FIG. 12 depicts a base 131 of the fuselage 25 that is located below the cabin floor 38 (FIG. 2)

on which a passenger may stand or sit. A bottom surface of the base **131** is coupled to a bracket **145**, referred to hereafter as “center bracket,” and a plurality of brackets **146**, referred to hereafter as “perimeter brackets,” positioned close to a perimeter of the base **131**. Each leg **36** is coupled to the center bracket **145** and at least one perimeter bracket **146**. Further, the center bracket **145** is coupled to an end of each leg **36** by an elastic coupling **149** that is designed to provide an adjustable length to permit rotation of the leg **36** to which it is coupled, as will be described in more detail below. In this regard, the elastic coupling **149** may be composed of an elastic material that permits the coupling **149** to stretch, or the elastic coupling **149** may be implemented with a spring or other type of device that permits the coupling **149** to stretch under tension. An exemplary elastic coupling **149** will be described in more detail below with reference to FIGS. **13A-D**.

(88) As shown by FIG. **12**, each leg **36** is coupled to a respective pair of perimeter brackets **146** in some embodiments. In this regard, for each perimeter bracket **146**, a pin **151** passes through the bracket **146** and the leg **36** to which it is coupled, and the leg **36** may pivot about the pin **151**. Note that the same pin **151** may pass through both perimeter brackets **146** coupled to the same leg **36**, or a separate pin **151** may be used for each perimeter bracket **146**.

(89) The brackets **145**, **146**, elastic coupling **149**, and pins **151** form a suspension system that permits the legs **36** to move relative to the fuselage **25** when they contact ground. In this regard, FIG. **13A** shows the landing gear prior to a leg **36'** making contact with the ground during landing. FIG. **13C** shows the landing gear after the leg **36'** makes contact with the ground causing the leg **36'** to move relative to the fuselage **25** and the other legs **36**, as shown by the separation of the leg **36'** from a reference plane **165**. To permit the leg **36'** to move upward, the leg **36'** pivots about the pin **151** that is used to couple it to the base **131**. Rotation about this pin **151** puts the elastic coupling **149** in tension such that the length of the coupling **149** is increased. However, the elastic coupling **149** is designed to resist this movement such that once the force applied from the ground is removed (e.g., after takeoff) a force generated by the elastic coupling **149** is sufficient to cause the leg **36'** to rotate back to its original position shown by FIG. **13A**.

(90) As best shown by FIG. **13B**, the elastic coupling **149** has a pair of rods **154**, **156** that pass through a can **153**. Positioned within the can **153** is an elastic bumper **157** through which an end of the rod **154** passes. The elastic bumper **157** is composed of an elastic material, such as rubber, which allows the bumper **157** to deform under an applied force, but the material of the bumper **154** resists such deformation and returns to its original shape when the applied force is removed.

(91) The opposite end of the rod **154** is positioned within a sleeve **160** of a rod end spherical bearing **161**, also referred to as a “Heim joint.” The bearing **161** has a pin **164** around which the sleeve **160** is permitted to rotate thereby accommodating slight angular changes between the rod **154** and the leg **36'** as the leg **36'** rotates.

(92) The can **153** also has a cap **158** through which the rod **156** passes. The opposite end of the rod **156** is positioned within a sleeve **170** of a rod end spherical bearing **172**. The bearing **172** has a pin **162** around which the sleeve **170** is permitted to rotate thereby accommodating slight angular changes between the rod **156** and the center bracket **145** as the leg **36'** rotates.

(93) Rotation of the leg **36'** relative to the fuselage **25** presses a head **173** (FIG. **1D**) of the rod **154** against the elastic bumper **157** causing the elastic bumper **157** to deform. Such deformation permits the rod **154** to move relative to the can **153** in a direction toward the rotating leg **36'** such that the overall length of the elastic coupling **149** is increased, as shown by FIG. **13D**. Thus, when the leg **36'** contacts the ground, the leg **36'** rotates relative to the fuselage **25** about the pin **151** to which it is coupled, thereby putting the elastic coupling **149** under tension such that its overall length is increased.

(94) Once the bumper **157** is fully deformed, further movement of the rod **154** relative to the can **153** is prevented thereby preventing further rotation of the leg **36'** relative to the fuselage **25**. In some embodiments, about half of an inch extension of the elastic coupling **149** translates into about three inches of movement of the end of the leg **36'**, but other ratios of movement are possible in

other embodiments.

(95) During takeoff, the force applied to the leg **36'** by the ground is reduced and eventually removed entirely. As this force is reduced, the deformation of the bumper **157** provides a force that tends to push the head **173** of the rod **154** away from the leg **36'** to which it is coupled so that the bumper **157** can return to its original shape, thereby decreasing the overall length of the elastic coupling **149**. This force is sufficiently strong to cause the leg **36'** to rotate about the pin **151** in the opposite direction as before such that the leg **36'** is rotated back to its original prior to the previous contact with the ground.

(96) Note that the other legs **36** may be similarly configured such that they also are permitted to rotate relative to the fuselage **25** when they make contact with the ground and during takeoff, as described above. This permitted movement of the legs **36** relative to the fuselage **25** has many benefits, such as providing for a smoother landing relative to an embodiment for which the legs **36** are rigidly mounted to the fuselage **25**. In addition, the permitted movement of the legs **36** may provide for safer landings for which the risk of toppling of the aircraft **20** is reduced by helping to accommodate for slight attitude misalignments of the fuselage **25** relative to the ground. This may be particularly advantageous for aircraft with relatively high aspect ratios of the fuselage or rotors, as described above, given that a higher fuselage or powerplane aspect ratio may have the effect of increasing the height of the aircraft's center of mass.

(97) As shown by FIGS. **14A** and **14B**, the distal end of each leg **36** away from the fuselage **25** may have a roller **142**. In one embodiment, the roller **142** is a self-castering roller mounted inside a large diameter, thin section bearing, and the roller **142** is capable of rolling in any direction (360 degrees) on the surface of the ground after landing, thereby helping to facilitate suspension function and ground handling. In other embodiments, other types of rollers and configurations of the legs **36** are possible. As an example, it is possible for the legs **36** not to have rollers or otherwise be configured for rolling.

(98) Referring to FIGS. **15A** and **15B**, the powerplane assembly **33** is connected to the powerpack assembly **32** by pivotal coupling **152**, which in some embodiments is implemented with a biaxial connector, though other types of couplings may be used in other embodiments. The pivotal coupling **152**, when implemented as a biaxial connector, functions like a U-joint and in this embodiment has a pair of pillow blocks **155** that are attached to a boom mount **163**, which is connected to the booms **51** as shown by FIG. **15A**. In this regard, bolts **166** or other couplers may be used to attach each boom **51** to the boom mount **163**, and bolts **167** (FIG. **16**) or other couplers may be used to attach the pillow blocks **155** to the boom mount **163**. As an example, each pillow block **155** may have holes **169** (FIG. **16**) through which a respective bolt **167** or other coupler may pass and extend into the boom mount **163** in order to secure the boom mount **163** to the pillow block **155**. In this embodiment, the boom mount **163** does not move relative to each pillow block **155**. FIG. **16** shows the pillow blocks **155** with bolts **167** inserted in one of the pillow blocks and the other pillow block **155** without bolts **167** inserted for illustrative purposes.

(99) Note that a biaxial connector generally refers to a connector that permits separate, independent rotation or combined, simultaneous rotation about two perpendicular axes. As an example, in the pivotal coupling **152**, the boom mount **163** and, thus, powerplane assembly **33** are permitted to pivot about one or more pins **183** for rotation about a first axis and to pivot about one or more pins **181** for rotation about a second axis, as will be described in more detail below. In other embodiments, other types of biaxial or non-biaxial couplings may be used to pivotally connect the powerplane assembly **33** to the fuselage **25**. As an example, hinges, spherical bearings or other types of mechanical couplings may be used to permit pitch and roll of the powerplane assembly **33**. In some embodiments, a compliant, flexible coupling may be used to connect the powerplane assembly **33** to the fuselage **25**. Such a coupling may comprise rubber, wire rope, webbing, or other type of compliant, flexible coupling to permit relative movement of the powerplane assembly **33** and the fuselage **25** about one or more axes. Yet other types of couplings are possible in other

embodiments.

(100) Note that one or more sensors **69** (FIG. **15A**), such as an inertial measurement unit, may be mounted on the boom mount **163** and/or other components of the powerplane assembly **33** for sensing the orientation (e.g., inclination) of the powerplane assembly **33**. As an example, one inclinometer may be attached to the boom mount **163** or other component of the powerplane assembly **33** to sense the angle of the powerplane assembly **33** relative to the roll axis, and another inclinometer may be attached to the boom mount **163** or other component of the powerplane assembly **33** to sense the angle of the powerplane assembly **33** relative to the pitch axis. In other examples, other types of sensors may be used to sense the orientation of the powerplane assembly **33** or make other types of measurements.

(101) As shown by FIGS. **16** and **17**, the pivotal coupling **152** has a yoke **175** that is rigidly mounted on the powerpack assembly **32** of the fuselage **25**. That is, the yoke **175** does not move relative to the powerpack assembly **32** thereby preventing the powerplane assembly **33** from rotating about the yaw axis (Z) relative to the fuselage **25**. Thus, the powerplane assembly **33** and the fuselage **25** rotate in unison about the yaw axis during flight.

(102) The yoke **175** comprises a pair of brackets **177** that protrude from a surface of the yoke **175** forming a space between the brackets **177**. The pillow blocks **155** are connected to each other by a cross **180** having a sleeve **182** and a pin **183**, which passes through the sleeve **182** and each bracket **177**. The sleeve **182** passes through the space between the brackets **177** and may rotate about the pin **183** thereby permitting the powerplane assembly **33**, including the rotors **52**, to rotate about the pitch axis.

(103) A respective pin **181** (which is hollow in one embodiment but may be solid) extends from each end of the sleeve **182** and passes through a pillow block **155**. In the embodiment depicted by FIGS. **16** and **17**, separate pins **181** extend from each end of the sleeve **182**. Each pillow block **155** may rotate about the respective pin **181** passing through it thereby permitting the powerplane assembly **33**, including the rotors **52**, to rotate about the roll axis. FIG. **18** shows a cross-sectional view of the pivotal coupling **152** along the cut lines depicted by FIG. **17**, and FIG. **19** shows a cross-sectional view of the pivotal coupling **152** along the cut lines depicted by FIG. **18**.

(104) Notably, the use of a pivotal coupling **152** that allows the fuselage **25** (e.g., the powerpack assembly **32** and the cabin **31**) to hang from a pivot point, as described herein, is relatively simple and much less expensive than other VTOL designs that use tilting rotors or tilting wings to generate lateral thrust. Indeed, many tilt rotor or tilt wing configurations require actuators and complex feedback circuits for actively moving a wing or rotor to a desired orientation relative to the fuselage. In the instant embodiment, differential thrust generated by the rotors **52** is used to pitch or roll the powerplane assembly **33** as desired, and there is no need for a separate actuator to control the orientation of the powerplane assembly **33** relative to the fuselage **25**. Instead, the fuselage **25** freely hangs from the pivotal coupling **152**, and gravity holds the fuselage **25** in an upright configuration while the powerplane assembly **33** is rotated about the pitch axis and roll axis via differential thrust generated by the rotors **52** as may be desired. That is, the powerplane assembly **33** rolls and pitches relative to the fuselage **25** when differential thrusts by the rotors **52** change the attitude of the powerplane assembly **33** while the fuselage **25** passively hangs from the pivotal coupling **152** in an upright orientation.

(105) Through the use of a pivotal coupling **152** as described above, it is possible to control the attitude of the powerplane assembly **33** in a manner conducive for optimizing flight while the fuselage **25** (e.g., the powerpack assembly **32** and the cabin **31**) essentially hangs from the pivotal coupling **152** with a substantially upright orientation. That is, pitching or rolling of the powerplane assembly **33** does not induce a corresponding movement of the cabin **31**. Thus, a passenger in the cabin **31** experiences much less pitch and roll movement as compared to aircraft designs which incline the airframe to generate lateral thrust. In addition, the pivotal coupling **152** acts as a mechanical filter to dampen out vibrations from the rotors **52** such that less vibration is felt in the

cabin **31**.

(106) Furthermore, by decoupling movement of the powerplane assembly **33** from the fuselage **25** about the pitch axis and the roll axis as described above, the powerplane assembly **33** is able to respond to rotor thrust changes more quickly since the weight and moment of inertia of the fuselage **25** does not resist the powerplane assembly's movement about the pitch axis and the roll axis. This enhances powerplane responsiveness and increases maneuverability of the aircraft **20**, which may be particularly important in urban areas where sudden or abrupt course changes might be needed or desired.

(107) FIG. **20** depicts an embodiment of the boom mount **163** connected to one boom **51** and the pillow blocks **155** of the pivotal coupling **152** (FIG. **16**). The boom mount **163** comprises a plurality of hollow brackets **201**, each of which has a shape corresponding to that of the proximal end of a boom **51** to be secured to the bracket **201**. In the embodiment depicted by FIG. **21**, the proximal end of each boom **51** is hollow for receiving a bracket **201** and has a rectangular cross-section, and the bracket **201** has walls that form the same approximate cross-sectional shape as the proximal end of the boom **51**. Thus, the proximal end of the boom **51** forms flat, faceted surfaces that are dimensioned to receive a respective bracket **201** of the boom mount **163**. In some embodiments, the proximal end of a boom **51** and the bracket **201** to be mated with the boom **51** are dimensioned such that the bracket **201** snugly fits into the proximal end of the boom **51**. Once the bracket **201** is inserted into the proximal end of the boom **51**, one or more bolts **166** or other types of couplers are inserted through the walls of the boom **51** and the bracket **201** to secure the boom **51** to the bracket **201**. As shown by FIG. **20**, each bracket **201** has one or more horizontal spindles **205** that span the inner walls of the bracket **201** to help stabilize and prevent deformation of the bracket walls.

(108) The distal end of each boom **51** is coupled to a respective rotor **52** and motor **66** (FIG. **2**). As shown by FIG. **21**, each rotor **52** has a plurality of rotor blades **225**, and its motor **66** is configured to operate under the control of a corresponding motor controller **65** (FIG. **4**) to rotate the blades **225** such that they interact with air to force air downward thereby providing a thrust in the opposite direction. As known in the art, the speed at which the blades **225** rotate or spin generally controls the amount of thrust provided by the rotor **52**. In one embodiment, the diameter of each rotor **52** is about 76 inches. That is, the distance from the center of the rotor **52** to the tip of a given blade **225** is about 38 inches. Further, the booms **51** are dimensioned such that the overall width (w.sub.p) of the rotors **52** of the powerplane assembly **33** is about 15.7 feet. That is, the maximum reach of the rotor blades **225** from the longitudinal axis of the fuselage **25** is about 7.85 feet. In other embodiments, other dimensions of the rotors **52** and booms **51** are possible.

(109) In addition, as previously noted above, the vertical orientation of the fuselage **25** with the powerplane assembly **33** mounted above the fuselage **25** helps to raise the rotors **52** off of the ground for safety while utilizing a relatively small footprint and/or aircraft size. For example, in the embodiment described above for FIG. **1A**, the rotors **52** may be located about 15.8 feet off of the ground, which is more than sufficient to ensure that they do not inadvertently contact a person walking fully erect beneath the rotors **52**. In this regard, keeping the rotors **52** about 10 feet or more off of the ground should make the height of the rotors **52** greater than the height of even the tallest person likely to use the aircraft **20** while adding a suitable margin of error. In addition, a greater height of the rotors **52** helps to reduce the effects of ground effect and potential for rotor wash recirculation interactions during takeoff and landing when the aircraft **20** is close to the ground. Indeed, such a height of the rotors **52** from the ground should position rotors **52** out of ground effect, thereby providing improved hover performance during landing and takeoff. Greater rotor height also reduces risk of injury to a passenger during an event that causes rotor or blade separation and also the risk of damage from foreign object debris (FOD) from objects on the ground that may be blown by the rotor wash. By creating a tall, narrow column of air, there should be reduced horizontal downwash velocity (increasing safety to passengers or other persons in the

vicinity of the aircraft **20**) and reduced risk of FOD recirculation as compared to aircraft designs with a shorter rotor height, such as at least some helicopters.

(110) In some embodiments, the aircraft **20** has an aspect ratio ($h_{sub.p}/w_{sub.p}$), referred to hereafter as “powerplane aspect ratio,” of about 1.0 or greater, where “ $h_{sub.p}$ ” is the maximum rotor height of the powerplane assembly **33** in the z-direction and “ $w_{sub.p}$ ” is the maximum rotor width (i.e., from blade tip-to-blade tip) of the entire powerplane assembly **33** in the x-direction, as shown by FIG. 3. As an example, in one embodiment described above, $h_{sub.p}$ is about 15.8 feet and $w_{sub.p}$ is about 15.7 feet. However, other dimensions and ratios are possible. Indeed, greater rotor heights may help to enhance at least some of the benefits described above, including reduced ground effect and potential for rotor wash recirculation interactions that can affect controllability of the aircraft **20** close to the ground, as well as enhancing safety and the passenger perception of safety. In fact, in some embodiments, it may be desirable for the powerplane aspect ratio to be at least 2.0 or even 3.0 or more.

(111) Note that the design of the aircraft **20** with a high powerplane aspect ratio helps to de-couple the ground footprint from rotor size. Thus, rotor diameter can be increased as desired to increase lift, reduce noise, or provide other benefits, without increasing the aircraft's ground footprint. Also, such design along with a streamlined fuselage **25** provides reduced vehicle interaction with rotor downwash, thereby providing for steady, smooth takeoffs and landings, particularly in conjunction with light, responsive powerplane differential thrusting. Various other benefits of a relatively high powerplane aspect ratio are also possible.

(112) Note that the use of electrical motors **66** in the configurations shown and/or described herein provide many advantages and benefits. In this regard, electrical motors **66** are clean and efficient helping to reduce pollution and waste. In addition, electrical motors **66** typically are much quieter than other types of motors, thereby enabling the aircraft **20** to access more locations in urban environments. Using multiple rotors **52** helps to reduce the required speed of the rotor blades of each respective rotor **52** in order for the powerplane assembly **33** to generate a desired amount of lift. In addition, using a vertically elongated fuselage **25** allows for a greater height of the powerplane assembly **33**, thereby permitting larger rotor blades to be used without increasing the aircraft's ground footprint. Using larger, slower-rotating blades decreases the amount of noise generated by the rotors **52** and, thus, helps to make the aircraft more conducive to urban environments.

(113) During operation, the flight controller **63** (FIG. 4) is configured to receive input from the sensors **69** and to control flight of the aircraft **20** based on the sensor inputs. As an example, the sensors **69** may indicate aircraft attitude and movement (e.g., airspeed), and the flight controller **63** may provide control signals to the motor controllers **65** for controlling the rotors **52** to generate the appropriate thrust for achieving a desired attitude of the powerplane assembly **33** and movement of the aircraft **20**. At least some sensors **69** (e.g., a GPS sensor) may be used to sense the location of the aircraft **20**, and the flight controller **63** may control movement of the aircraft **20** in order to navigate the aircraft **20** to a desired destination. At least some sensors **69**, such as radar or LIDAR, may be used to sense objects external to the aircraft **20**, and the flight controller **63** may use information from these types of sensors to sense and avoid collision threats.

(114) Based on the input from the sensors **69**, the thrusts generated by the rotors **52** are controlled to move the aircraft **20** as desired to a destination. As an example, thrust may be increased to lift the aircraft **20** off the ground. After taking off, differential thrusting may be used to control the direction of flight and attitude of the powerplane assembly **33**. Thrusting may also be used to control the ascent and descent of the aircraft **20** as may be desired. As the attitude of the powerplane assembly **69** is changing about the pitch axis and roll axis, the fuselage **25** hangs from the pivotal coupling **152** and maintains a vertical orientation due to gravity independent of the movements of the powerplane assembly **33** and, thus, rotors **52** about the pitch axis and roll axis.

(115) Once the aircraft **20** reaches its destination, the flight controller **63** controls the rotors **52** so

that the aircraft **20** gently lands on the legs **36**. If an emergency condition is detected during the flight, such as a detection of a fire or failure of a critical component (e.g., one or more rotors **52**), the flight controller **63** may be configured to actuate the BRS **72** in order to deploy a parachute that allows the aircraft **20** to slowly and safely return to the ground. In other embodiments, other techniques and/or recovery systems for handling an emergency condition are possible.

(116) FIG. **22** shows a VTOL aircraft **200**, according to an example embodiment. The aircraft **200** has a fuselage **250** that is vertically elongated (in the z-direction). As an example, the fuselage **250** may be vertically elongated such that it has an aspect ratio ($h_{\text{sub.f}}/w_{\text{sub.f}}$), referred to hereafter as “fuselage aspect ratio,” of about 2.0 or greater, where “ $h_{\text{sub.f}}$ ” is the maximum height of the fuselage **25** in the z-direction and “ $w_{\text{sub.f}}$ ” is the maximum width of the fuselage **25**. However, a greater fuselage aspect ratio may help to increase the height of the powerplane assembly **330**, thereby realizing various benefits, as will be described in more detail below. To enhance at least some of these benefits, it may be desirable for the fuselage aspect ratio to be at least 3.0 or even 4.0 or more. For example, a greater fuselage aspect ratio may help to minimize or eliminate downwash ground effect interactions, thus simplifying control system workload and enhancing passenger comfort. Having a greater rotor height not only provides some performance benefits, as described in more detail below, but also may help to provide passengers with a greater sense of safety. It may also reduce the rotor wash felt by a passenger or other user while walking beneath the rotors **520**.

(117) The vertical orientation of the fuselage **250** also helps to optimize the shape of the fuselage **250** for streamwise aerodynamic efficiency and reduced VTOL and hover drag. Some additional benefits include: providing more room to accommodate one or more standing passengers and providing easy entry and exit to the cabin for passengers; reduced ground footprint, helping to facilitate maneuvers in confined or congested areas; and providing greater vertical cabin space to accommodate long, unwieldy cargo. In other embodiments, aspect ratios below 2.0 are possible, and it is unnecessary for the fuselage **250** to be vertically elongated. Indeed, although the vertical orientation of the fuselage **250** has many advantages, as described herein, it should be emphasized that such vertical orientation is unnecessary in all embodiments, and any shape of the fuselage **250** is possible in other embodiments.

(118) As shown in FIG. **22**, the fuselage **250** has one or more modules, such as a cabin **310**, a powerpack assembly **320**, a powerplane section **350**, and a nose **340** (which is tapered in the shape of a cone, though other shapes are possible). The fuselage **250** is coupled to a powerplane assembly **330**, which includes booms **510** and rotors **520**, used to provide lift and controllability as will be described in more detail below. In some embodiments, the walls of the fuselage **250** are composed of a carbon fiber material helping to reduce weight. As shown by FIG. **22**, the aircraft **200** lacks the landing gear of aircraft **20**, because aircraft **200** is configured to land and dock in a respective VTOL docking station **900**, as described in more detail below.

(119) In the embodiment depicted by FIG. **22**, the aircraft **200** is dimensioned such that, when it is on the ground prior to takeoff, the bottom of the cabin **310** is about 1.2 feet above the ground, the bottom of the powerpack assembly **320** is about 9.1 feet above the ground, the bottom of the powerplane section **350** is about 16.6 feet above the ground, and the top of the nose **340** is about 18.1 feet above the ground such that the overall height of the fuselage **250** is about 18.1 feet above the ground.

(120) In some embodiments, the aircraft **200** is designed to have a cruise speed of up to about 60 miles per hour with a weight of about 900 pounds and a targeted service range of about 10 miles (with reserve). In addition, the aircraft **200** is designed to produce noise during takeoff and landing of less than 60 decibels (dB) from a distance of 50 feet. In other embodiments, other design and performance characteristics are possible.

(121) The cabin **310** forms a compartment in which cargo, passengers, and/or other types of payloads may be situated for transport, and will sometimes be referred to hereafter as “passenger cabin.” In the embodiment depicted by FIG. **22**, the cabin **310** is designed to accommodate one or

two passengers, who stand on a floor **380** (FIG. 22) of the cabin **310** during operation, similar to an elevator or some alpine cable cars. Also, the cabin **310** has a door **410** through which the passenger may enter the cabin **310**, and the cabin **310** has a plurality of windows **430** through which the passenger may see outside the cabin **310**. In other embodiments, the cabin **310** may be designed to accommodate a greater number of passengers and/or other types of payloads, such as cargo (e.g., one or more packages to be delivered to one or more locations), and use of windows **430** is unnecessary. In addition, the cabin **310** may include one or more seats for permitting one or more passengers to sit as may be desired during transport. In contrast to door **41**, door **410** can be configured as an automatic, single sliding plug door, as is common in the light rail industry instead of a hinged door. Door **410** can be desirable when aircraft **200** is docked within a docking station **900** (described below), because a hinged door that swings outward such as door **41** may interfere with docking station **900**. In addition, single sliding plug door design and engineering, including tamper-proof automatic locking mechanisms for safety-critical, high-speed applications, is mature technology well understood within the industry.

(122) The vertical orientation of the fuselage **250** and cabin **310** allows a passenger to stand or maintain an upright posture, providing a user experience similar to an elevator and which may be preferable or more enjoyable to the at least one passenger, enables rapid entry and exit, and is easy to clean between flights. Further, as will be described in more detail below, the aircraft **200** may be designed such that the cabin **310** experiences very little or no movement about the pitch and roll axes in response to differential thrusting of the rotors **520**, thereby reducing perturbations or movements experienced by the passenger during flight. It should be understood that aircraft **200** is compatible with the BRS system discussed with respect to aircraft **20**, the full description of which is omitted herein for brevity. The BRS may be located above the powerplane assembly **330**, helping to keep the parachute lines clear of the powerplane rotors **520**. However, in other embodiments, aircraft **200** may use non-BRS flight safety and recovery systems. For example, aircraft **200** may utilize a dual redundant battery and motor system such that if the if one set of batteries or motors fail, the backup batteries or motors can be utilized to safely land aircraft **200**.

(123) In some embodiments, the shape of the vertically elongated fuselage **250** is streamlined so as to reduce aerodynamic drag from air being blown over the fuselage **250** by the rotors **520**. In the embodiment depicted by FIG. 22, the fuselage **250** has a generally circular cross-section along its longitudinal axis (which is parallel to the z-direction in the exemplary embodiment of FIG. 22), and the fuselage **250** may be tapered. As an example, and as shown by FIG. 22, the fuselage's diameter or width may increase downward in the z-direction from the top of the fuselage **250** in order to provide greater area for the powerpack assembly **320** and the cabin **310**. However, as shown in FIG. 22, the diameter of the lower portion of the cabin **310** may slightly decrease toward the base of the cabin **310** in order to reduce frontal area and aerodynamic resistance. In some embodiments, the maximum diameter or width of the cabin **310** is about 38 inches, but other maximum cabin widths are possible. In other embodiments, the fuselage **250** may have other types of profiles and contours, and as noted above, the pivotally-coupled powerplane assembly described herein is compatible with a wide range of vertically or horizontally elongated fuselage configurations.

(124) The shape and dimensions of the cabin **310** are designed such that at least one passenger may comfortably stand fully erect in the cabin **310** during flight. In some embodiments, the interior height of the cabin **310** from floor to ceiling is about 7.9 feet, though other heights are possible in other embodiments. Indeed, cabin heights greater than 6.5 feet will accommodate 99th percentile males in order to allow them to stand fully erect in the cabin **310**. In some embodiments, the cabin **310** may have one or more holding apparatuses, such as a handrail (not shown) or hand strap, mounted on an interior wall of the cabin **310**, and a passenger may hold onto such holding apparatus during flight in order to help maintain balance. To provide more room for standing, the cabin **310** may be designed without a passenger seat, but it is possible for the cabin **310** to have one or more seats in which one or more passengers may sit during flight. Yet other designs and

configurations of the cabin **310** are possible in other embodiments.

(125) As shown in FIG. **22**, the powerpack assembly **320** of aircraft **200** is positioned above cabin **310** and below powerplane section **350**. The positioning of the powerpack assembly is not arbitrary. Placing the powerpack mass close to the powerplane pivot (e.g., pivotal coupling **1520**) helps maintain cabin verticality during lateral acceleration, because multirotor aircraft pivot around the center of mass, not the center of thrust. Accordingly, positioning the powerpack assembly **320** near the pivotal coupling **1520** helps aircraft **200** to maintain cabin verticality during lateral acceleration as shown in FIG. **35D**. By contrast, multirotor aircraft with a center of mass located far below the powerplane pivot may have difficulty maintaining vertical orientation of the cabin during pronounced lateral acceleration, as illustrated in FIGS. **35A-C**.

(126) Returning to FIG. **22**, the powerplane assembly **330** comprises a plurality of booms **510** extending from a central mount and a plurality of motors **660** and rotors **520** respectively mounted on the booms **510**. In the embodiment depicted by FIG. **22**, the powerplane assembly **330** has four booms **510** and four rotors **520** with five blades each, but the powerplane assembly **330** may have other numbers of booms **510**, motors **660**, rotors **520**, or rotor blades **225** in other embodiments. In the embodiment depicted by FIG. **22**, the rotors **520** provide sufficient lift for the aircraft **200** without the need of other lift-generating apparatuses, such as wings. However, in other embodiments, the aircraft **200** may have winglets or other lift-generating apparatuses, which may help to increase the range of the aircraft **200**.

(127) The rotors **520** are driven by motors **660** that cause the rotors **520** to generate aerodynamic thrust to be used for generating lift and controlling powerplane pitch, roll, and yaw. In this regard, FIG. **24** shows an exemplary flight control system **600** of the aircraft **200** that can be used to control movement of the aircraft **200**. In the embodiment depicted by FIG. **24**, the system **600** comprises a flight controller **63** that is coupled to a plurality of motor controllers **65**, which are configured to control motors **660** for driving the rotors **520**. The flight controller **63** is also coupled to a plurality of sensors **69** that are configured to sense flight characteristics of the aircraft **200**, such as horizontal airspeed, vertical airspeed, heading, powerplane attitude, and other flight characteristics that are typically used to control flight of conventional aircraft. The sensors **69** may also sense other parameters useful in controlling the aircraft **200**, such as external objects for sensing and avoiding collision threats, temperature, etc.

(128) The flight controller **63** is configured to control the general operation of the aircraft, including flight, as will be described in more detail below. As an example, the flight controller **63** may be configured to provide control inputs to the motor controllers **65** for controlling rotors **520** such that they generate thrust as appropriate for controlling the movement of the aircraft **200** in a desired manner. That is, the flight controller **63** instructs each motor controller **65** to operate its respective rotor **520** at a desired speed so that appropriate thrusts are generated for controlling the flight of the aircraft **200**. As an example, the rotors **520** may be controlled to generate sufficient lift to counteract the weight of the aircraft **200** such that the aircraft **200** is flown to a desired altitude above ground. Further, differential thrusting of the rotors **520** may be used to control movement of the powerplane assembly **330** about the roll and pitch axes, as is known in the art for conventional multirotor VTOL aircraft, and also to provide horizontal movement of the aircraft **200** as may be desired. Further, differential torque from the rotors **520** may be used to control movement of the powerplane assembly **330** about the yaw axis, as is known in the art for conventional multirotor VTOL aircraft.

(129) In addition, in some embodiments, as will be described in more detail below, the powerplane assembly **330** may be coupled to the fuselage **250** via a pivotal coupling **1520** (FIG. **31**) that permits the powerplane assembly **330** to pivot about at least two axes (e.g., pitch and roll) relative to the fuselage **250**. In such an embodiment, the differential thrusting used to rotate the powerplane assembly **330** about the pitch and roll axes does not cause corresponding movement of the fuselage **250**. That is, in some embodiments for which the pivotal coupling **1520** couples the powerplane

assembly **330** to the fuselage **250**, the powerplane assembly **330** pitches and rolls independently of the fuselage **250**. However, the powerplane assembly **330** may be indexed to the fuselage **250** such that the powerplane assembly **330** and fuselage **250** rotate together about the yaw axis. An exemplary pivotal coupling **1520** for achieving this effect will be described in more detail below with reference to FIG. **31**. However, it should be noted that use of such a pivotal coupling **1520** is unnecessary, and other types of couplings are possible in other embodiments.

(130) The flight controller **63** is further coupled to a power source **75**, which may include one or more batteries for powering the electrical motors and components of the aircraft **200**, and a ballistic recovery system (BRS) **72**. The power source **75** is also coupled to a power source controller **77**, which is configured to monitor power source for charge status, battery health, battery temperature, etc. Power source controller **77** is also configured to control the charging and discharging of the power source **75**. For example, as discussed below with respect to FIG. **25**, the power source controller **77** can be configured to communicate with a docking station controller **902** of a VTOL docking station **900** in order to control the charging characteristics of the power source **75** when the aircraft **200** (described below with respect to FIGS. **22-30**) docks on the VTOL docking station **900**.

(131) Returning to FIG. **24**, the flight controller **63** and the power source controller **77** may be implemented in hardware or a combination of hardware with software and/or firmware. As an example, the flight controller **63** and the power source controller **77** may be implemented with hardware gate logic, such as a field programmable gate array (FPGA), or with application-specific integrated circuits (ASICs). In some embodiments, the flight controller **63** and/or the power source controller **77** each comprise at least one processor, such as a digital signal processor (DSP) or a central processor unit (CPU), that is configured to execute software and/or firmware for performing the functions described herein. In other embodiments, other types of controllers **63** and/or power source controllers **77** may be used.

(132) In some embodiments, the motors **660** are electrical and draw electrical power from the power source **75** during operation. The motors **660** may operate under the control of one or more motor controllers **65** to control the speeds of the rotors **520** as instructed by the flight controller **63**. In some embodiments, the aircraft **200** is autonomous, and the flight controller **63** controls flight of the vehicle **200**, including both navigation and aviation, without the use of or inputs from a pilot. However, in other embodiments, a pilot onboard the aircraft **200** may provide control inputs for controlling flight. Such inputs may be received by the flight controller **63**, which interprets such inputs to provide appropriate control signals to the motor controllers **650**. If desired, such inputs may be received from a remote pilot via wireless communication. Yet other techniques and configurations for controlling the flight of the aircraft **200** are possible in other embodiments. It is also possible to drive the rotors **520** with other types of motors.

(133) The powerpack assembly **320** (FIG. **22**) houses the power source **75** (FIG. **24**) and motor controllers **65**, similar to FIGS. **8A** and **8B** with respect to powerpack assembly **32**. As discussed with respect to powerpack assembly **32**, in one embodiment, the power source **75** of powerpack assembly **320** comprises a plurality of battery packs **80** that are positioned just below the motor controllers **65**, as shown by FIGS. **8B** and **8C**. In other embodiments, other types of power sources **75** may be used, and the relative positions of the battery packs **80** and the motor controllers **65** may be different. In some embodiments, as depicted with respect to powerpack assembly **32** in FIGS. **8A**, **8B**, **8C**, and **22**, the powerpack assembly **320** may be located below the powerplane assembly **330**, including the motors **660** and rotors **520**. By locating the battery packs **80** and motor controllers **65** just below the powerplane assembly **330**, the lengths of electrical wiring between the battery packs **80**, motor controllers **65**, and rotor motors **660** can be kept relatively short, thereby helping to minimize the electrical resistance and weight of such wiring. Additionally, locating the battery packs **80** and motor controllers **65** just below the powerplane assembly **330** helps maintain cabin verticality during lateral acceleration, as is illustrated in FIGS. **35A-D**. In some

embodiments, the modular powerpack assembly **320**, which comprises the battery packs **80**, the motor controllers **65**, and the flight controller **63**, can be removed from the fuselage **250** and replaced with a new powerpack assembly **320**.

(134) In one embodiment, the powerpack assembly **320** for aircraft **200** comprises 1012 battery cells providing a total power of approximately 99 kilowatt hours (kWh), although other numbers of cells and amounts of available power are possible. These cells can be arranged in several interconnected battery packs **80** where each pack **80** comprises a plurality of cells. Similar to the embodiment shown by FIG. **8C**, the battery packs **80** can be arranged into two vertically-stacked towers **79** within powerpack assembly **320**. The battery packs **80** may be stacked in other arrangements, and other numbers of battery pack towers **79** may be used in other embodiments.

(135) In some embodiments, each battery pack tower **79** is electrically connected to each motor **660** for driving the rotors **520** so that each motor **660** may receive power from either or both of the towers **79**. In addition, each battery pack tower **79** is able to generate sufficient power to drive simultaneously all of the motors **660** without power from the other tower **79**, if desired. Thus, in the event of an operational problem with one of the towers **79**, the other tower **79** may provide sufficient power for all of the motors **660** for at least enough time for the aircraft **200** to continue to the destination or perform an emergency landing. However, in other embodiments, other types and arrangements of batteries may be used.

(136) As shown by FIG. **22**, an intake duct **820** is located in the powerpack assembly **320** of the fuselage **250** to receive cooling air from the rotors **520**. The air travels from this intake duct **820** through the powerpack assembly **320** and exits the powerpack assembly **320** via an outlet duct **850** that is located on an opposite side of the fuselage **250** relative to the intake duct **820**. The outlet duct **850** may be fed by a cooling fan, creating negative pressure within the powerpack assembly **320** to aid the intake flow. As the air is passing through the powerpack assembly **320**, it absorbs heat from the battery packs **80** and other electronics, and then is expelled through the outlet duct **850**, thereby ensuring the electronic equipment is maintained at its designed operating temperature.

(137) In some embodiments, the powerpack assembly **320** can be cooled via liquid cooling. For example, the fuselage **250** can be equipped with a heat exchanger (not shown) filled with a cooling liquid that circulates the cooling liquid through liquid channels disposed within the powerpack assembly **320**. In such embodiments, air traveling from the intake duct **820** can pass through the heat exchanger and out of the fuselage **250** via outlet duct **850**. As the cooling liquid circulates through powerpack assembly **320**, heat is drawn into the cooling liquid and is subsequently dispersed into the surrounding environment as the cooling liquid passes through the heat exchanger, in which the cooling liquid is cooled by the air traveling between intake duct **820** and outlet duct **850**. In some embodiments, the heat exchanger may additionally be equipped with a fan in order to increase the rate of heat exchange between the cooling liquid within the heat exchanger and the air of the surrounding environment, for example during charging of battery pack **80**.

(138) In the event of a fire in the powerpack assembly **320**, the vertical orientation of the fuselage **250** ensures that the passenger cabin **310** is well positioned below the smoke, heat or flame that may rise from such a fire. Additionally, the passenger cabin **310** is separated from the battery packs **80** by two bulkheads, each of which may be designed according to applicable aviation fire-resistance regulations, in order to ensure that the aircraft **200** reaches the ground and the passenger safely exits before the fire is able to penetrate the cabin **310**. For example, from an altitude of 500 feet above ground level (AGL) with a BRS descent rate of 10 fps, the fire-resistant bulkhead system would need to remain intact for at least 50 seconds, well within the capabilities of modern fire-resistant materials.

(139) Notably, aircraft **200** shown in FIG. **22** lacks landing gear. Instead of utilizing landing gear as was discussed with respect to aircraft **20**, aircraft **200** is configured to dock with a respective VTOL docking station **900**. An exemplary VTOL docking station **900** is shown in FIG. **23**. Turning to FIG. **25**, aircraft **200** is synergistically designed to dock with VTOL docking station **900** by resting

the lower surface of each structural boom **510** on the upper rim of VTOL docking station **900**, effectively suspending the aircraft **200** from its booms **510** and the frame **906** of the VTOL docking station **900**. However, it should be noted that in some embodiments, aircraft **200** can be configured to operate without use of a VTOL docking station **900**, and may hover near the ground to allow passengers to board and deboard the aircraft **200** without requiring the aircraft **200** to dock with a docking station **900**, as will be described further with respect to FIG. **34**.

(140) FIG. **26** is an exploded view of VTOL docking station **900** showing the primary components including the base **918**, a barrier **908**, a structural frame **906**, an automatic door **910**, a door open and close mechanism (not shown), an access control system **912**, electrical charging components **916**, electrical contact strips **904**, a controller **902**, and lighting components **914** as desired for nighttime visibility, safety, aesthetics, or visually communicating system status of the VTOL docking station **900**. The VTOL docking station **900** is self-enclosed and fully autonomous, and provides a secure take-off and landing zone for aircraft **200** while also providing pedestrian protection, a small footprint for minimal site impact, and a fixed, defined take-off and landing location to assist autonomous vehicle navigation. Additional benefits of the VTOL docking station **900** include: a prefabricated construction to enable mass production cost savings; minimal site preparation and installation labor requirements; multi-jurisdictional regulatory code compliance; construction of high-grade industrial materials; well suited to wide distribution and rapid scaling of commercial operations; all weather, all season operation, safe harbor for aircraft **200** during storms; autonomous charging of aircraft **200**; and suitability for both indoor and outdoor installation.

(141) A plurality of VTOL docking stations **900** are configured to be conveniently placed throughout the environment in which aircraft **200** operates, and passengers may select from a range of journeys depending on their desired VTOL docking station **900** departure and destination locations.

(142) In some embodiments, the VTOL docking station **900** may be configured to have an approximately cylindrical shape roughly corresponding to the generally circular cross-section of aircraft **200**'s fuselage **250**, although in other embodiments, other shapes of VTOL docking station **900** are envisioned.

(143) The base **918** of VTOL docking station **900** is a self-contained, hollow structural container made of metal, fiber-reinforced composite, or other suitable material with integrated mounting points for the structural frame **906** and barrier **908**. The base **918** also provides a protective housing and mounting points for the VTOL docking station **900** subcomponents including the electrical charging components **916**, control and navigation support elements, the automatic door mechanisms for opening, closing, and locking, and the precision track upon which the door is configured to slide during operation.

(144) In the example embodiment of base **918** shown in FIG. **25** and FIG. **26**, the base **918** may be installed outdoors in a prepared hole in the ground, or indoors in a prepared opening in the floor, in such a way that the flat, upper surface of base **918** (e.g., the automatic door **910** entryway floor) is at roughly the same elevation as the surrounding ground or floor. Additionally, the upper surface of base **918** contains a recess for the bottom of cabin **310**, so that when aircraft **200** is docked (hanging at rest) in VTOL docking station **900**, the aircraft cabin floor **380** is approximately the same elevation as the flat, upper surface of base **918** (entryway floor) and the surrounding ground or floor.

(145) VTOL docking station **900** frame **906** may be constructed out of any structurally appropriate and suitably rugged and durable material such as aluminum, steel, stainless steel, fiber-reinforced composite, and/or other materials or combinations thereof, bearing in mind that frame **906** is intended to support the combined aircraft and payload with an appropriate factor of safety. The VTOL docking station **900** additionally includes a barrier **908** which surrounds the cabin **310** of aircraft **200** when aircraft **200** is docked within the VTOL docking station **900**. The barrier **908** may be constructed of any suitable material; in the example embodiment depicted in FIG. **26**, barrier

908 is constructed out of a transparent structural laminated glass material. In some embodiments, the VTOL docking station **900** may lack frame **906** and the barrier **908** can be configured to support the combined aircraft and payload with an appropriate factor of safety without the utilization of frame **906**.

(146) Access to and from the aircraft **200** is enabled by the automatic door **910** of VTOL docking station **900**, which may include access control **912**. Access control **912** may be configured to selectively provide access to the passenger requesting a ride from aircraft **200**, and may be implemented using a variety of known methods. For example, access control may be implemented as an RFID device that is configured to open in response to an appropriate RFID signal received from a user device associated with the passenger. In other embodiments, access control **912** may be implemented as a smart lock that can communicate wirelessly with a user device associated with the passenger. Access control **912** may be selectively tailored for arriving or departing passengers. For example, for a departing passenger access control **912** may prevent the automatic door **910** of the VTOL docking station **900** from opening until a charging cycle is complete and the vehicle has passed self-check and communicated flight readiness to the VTOL docking station controller **902**. After departing passenger boarding, access control **912** may be configured to close and lock aircraft cabin door **410** and VTOL docking station automatic door **910** in preparation for initiating aircraft **200** take-off procedures.

(147) For an arriving passenger, access control **912** may provide an enhanced level of safety by preventing the aircraft cabin door **410** and VTOL docking station automatic door **910** from opening until access control **912** receives an indication from the aircraft **200** (e.g., via communication between docking station controller **902** and flight controller **63**) that the rotors **520** of aircraft **200** have reduced thrust and/or rotation speed to below a predetermined value, thereby preventing passenger de-boarding until rotor wash and flying debris from the spinning rotors has subsided.

(148) Access control **912** may also prevent unauthorized access into an aircraft **200** that is docked with a VTOL docking station **900**. For example, access control **912** may be configured to allow automatic door **910** to open only after receiving a command from a user device associated with a passenger scheduled to utilize aircraft **200**, thereby preventing an unauthorized entry into aircraft **200** by another person who is not the scheduled passenger. Access control **912** can also prevent passengers and other persons from entering VTOL docking station **900** while no aircraft **200** is docked within, thereby ensuring an unobstructed landing area when an arriving aircraft **200** initiates a docking procedure. Many other access control **912** configurations may be implemented as desired in order to support safe and efficient operation of VTOL docking station **900**.

(149) The VTOL docking station controller may be configured to communicate with components of aircraft **200**. The controller **902** may be implemented in hardware or a combination of hardware with software and/or firmware. As an example, the VTOL docking station controller **902** may be implemented with hardware gate logic, such as a field programmable gate array (FPGA), or with application-specific integrated circuits (ASICs). In some embodiments, the VTOL docking station controller **902** comprises at least one processor, such as a digital signal processor (DSP) or a central processor unit (CPU), that is configured to execute software and/or firmware for performing the functions described herein. In other embodiments, other types of controllers **902** may be used.

(150) The VTOL docking station controller **902** may be configured to communicate with flight controller **63** of aircraft **200** to provide aircraft navigation support. While each VTOL docking station **900** has a fixed, known location and aircraft **200** uses autonomous navigation to follow predetermined flight paths between distributed VTOL docking stations **900** or other departure and destination points, precision landing of aircraft **200** in the VTOL docking station **900** may require enhanced, centimeter level positional accuracy. Therefore, VTOL docking station **900** may be equipped with a local-area navigation augmentation system to establish direct communication between VTOL docking station controller **902** and aircraft flight controller **63** as the aircraft **200** approaches the docking station **900**, and to provide high precision positional information and

navigation assistance to help guide aircraft **200** during its descent from hover into the VTOL docking station **900**. It should be understood that base controller **902** is intended to represent the entire navigation augmentation capability of the VTOL docking station **900**, which may encompass multiple transmitter or receiver units of various designs and operational frequencies, together with a range of hardware and software systems to provide high precision, all-weather, local-area navigation support during aircraft **200** take-off and landing.

(151) In some embodiments, VTOL docking station **900** may be equipped with various cargo-centric features. For example, in some embodiments, aircraft **200** may be configured to carry cargo in place of passengers, and docking stations **900** can be modified with cargo-centric features such as automated loading and unloading systems to transfer cargo between aircraft **200** and an integrated, secure cargo hub locker (not shown). Yet other cargo-centric features of docking station **900** are envisioned.

(152) The VTOL docking station **900** is equipped with an autonomous charging capability, so that docked aircraft **200** can be re-charged as necessary without human action. As shown in FIG. **26**, the main electrical charging components **916** are located within the base **918** of VTOL docking station **900** and are controlled by VTOL docking station controller **902** to charge power source **75** of a docked aircraft **200**. Also shown in FIG. **26** and FIG. **27B** are VTOL docking station electrical contact strips **904**, which are shown as curved in this embodiment, but may take any desired shape in other embodiments. The electrical contact strips **904** are positioned at the top of the VTOL docking station frame **906** and electrically coupled to the charging components **916** located in the base **918**.

(153) Turning now to the aircraft **200**, FIG. **27A** depicts powerplane assembly **330**, including boom mount **1630**, booms **510**, and pivotable coupling **1520**. Similar boom electrical contact strips **512** are mounted to the bottom of each boom **510**. In the embodiment depicted in FIG. **27A**, the boom electrical contact strips **512** are straight but can take on other shapes and dimensions in other embodiments as desired. It should also be understood that in some embodiments, each boom can include more than one boom electrical contact strips **512**, as desired. The boom electrical contact strips **512** are electrically coupled to the power source **75** of aircraft **200** and are used to charge aircraft **200** when aircraft **200** is docked with VTOL docking station **900**. Thus, each of the one or more VTOL docking station electrical contact strips **904** is configured to make contact with a respective boom electrical contact strip **512** when aircraft **200** docks with docking station **900**. The respective contact strips and their mating surfaces may be appropriately designed to handle high current, which together with a point load clamping force sufficient to ensure that good electrical contact is made, may support Direct Current Fast charging of aircraft **200** power source **75**.

(154) Advantageously, this unique plugless charging solution comprised of crossed electrical contact strips effectively de-tolerances vehicle position, and as illustrated in FIG. **27C**, the system can therefore accommodate charging of aircraft **200** in a wide range of vehicle x/y positions and z clocking angles within the VTOL docking station **900**, without human action. As shown in this simplified plan view, the aircraft **200** can be positioned at point P1 within docking station **900** (approximating the center of docking station **900**) and may also be shifted left or right along the x direction to points P2 and P3 without losing electrical connectivity between boom electrical contact strips **512** and the one or more electrical docking station electrical contact strips **904**. Similarly, aircraft **200** may also be shifted along the y direction in a similar manner as described with respect to P2 and P3, or rotated about its Z axis as shown for P1, or multiple combinations of X and Y displacement, with or without Z rotation, without losing electrical connectivity between boom electrical contact strips **512** and the one or more docking station electrical contact strips **904**.

According to some embodiments, each of the one or more docking station electrical contact strips **904** can extend in an arc angle of up to approximately 60 degrees. In other embodiments, docking station electrical contact strips **904** can extend in other arc angles as desired.

(155) In some embodiments, the number of docking station electrical contact strips **904** matches

the number of booms of aircraft **200**. For example, in the embodiment disclosed in FIG. **27B** there are four docking station electrical contact strips **904A-904D** that each extend over an approximate 60-degree arc, and in FIG. **27A** there are four matching boom electrical contact strips **512**. In other aircraft **200** and VTOL docking station **900** embodiments, there may be other numbers of booms **510**, boom electrical contact strips **512**, and docking station electrical contact strips **904**, and the number of electrical contact strips between the boom and docking station need not match.

(156) As previously mentioned, the VTOL docking station controller **902** may be configured to communicate with components of aircraft **200** including power source controller **77** of aircraft **200** in order to control aspects of charging the aircraft power source **75**. For example, docking station controller **902** may communicate with power source controller **77** to receive power source data that includes charge status of the power source **75**, temperature of the power source **75**, battery health of power source **75**, or many other parameters. Data received from power source controller **77** allows docking station controller **902** and the charging algorithm to control the electrical charging components **916** so as to safely and optimally charge the power source **75** while the aircraft **200** is docked with the VTOL docking station **900**. In some embodiments, communication between docking station controller **902** and power source controller **77** may be enabled wirelessly, and in other embodiments communication may be achieved by means of an optical or physical connection.

(157) In some embodiments, parallel charging of multiple power sources **75** (e.g., multiple battery packs) is possible. In some embodiments, aircraft **200** is not charged after every flight. For example, if the power source controller **77** determines that the power source **75** is beyond a threshold of charge (e.g., 80% capacity), power source controller **77** can instruct docking station controller **902** to not provide charge to the power source **75**. This can be advantageous, for example, to maximize battery longevity by avoiding unnecessary charge/discharge cycles of power source **75**.

(158) FIG. **28** depicts a structural boom **510** of aircraft **200** depicted in FIG. **22**. The structural booms of aircraft **20** and **200** are essentially cantilevered beams with an end mass consisting of motors **66** and **660** and rotor assemblies **52** and **520** assembly, respectively, and as such are structural members designed to resist lateral loads acting transversely to the beam axis. Boom **510** features a mechanically efficient thin wall composite laminate with a non-uniform diamond profile that offers improved stiffness (resistance to bending), a higher natural vibration frequency (first harmonic) with reduced amplitude, and improved aeroacoustics over boom **51** of aircraft **20**. Other materials or construction methods are possible, including traditional aluminum construction common in the aerospace industry, with varying benefit.

(159) Notably, boom **510** has an approximately diamond shape profile as compared to the rectangular profile of boom **51** as can be seen in FIGS. **28B-28C**. The diamond profile provides enhanced mechanical efficiency by placing the primary vertical and lateral bending load paths as far as possible from the neutral axis of the beam, maximizing the area moment of inertia for improved mechanical efficiency. Additionally, boom **510** connects to boom mount **1630** by means of fasteners thru load-bearing inserts **1660** which are placed directly in-line with the vertical and lateral bending load paths at the 'points' of the approximately diamond profile, significantly increasing fastener reaction mechanical efficiency compared with boom **51**. Further, the approximately diamond profile of boom **510** is taller than the rectangular profile of boom **51**, and because the load-bearing inserts **1660** are spaced at the diamond profile extremities rather than along the rectangular sidewalls as in boom **51**, the distance between the upper and lower fasteners of boom **510** is considerably greater than boom **51**, providing improved leverage to react the primary vertical bending moment of boom **510** compared with the more closely spaced fasteners and more circuitous reaction load path of boom **51**.

(160) Additionally, the approximately diamond profile of boom **510** is non-uniform: the cross-sectional height and width dimensions diminish along the boom 'X' axis shown in FIG. **28A**, from

the boom mount end to the rotor end, producing a double taper effect (e.g., along the illustrated y and z axes). Tapered cantilevered beams are well known to those with skill in the art for their high-stiffness-to-mass ratio and reduced bending moment and deflection compared with constant profile cantilevered beams. Finally, the approximately diamond profile of boom **510** may reduce undesirable acoustic effects by minimizing and disbursing reflections of the high-pressure wave below the passing rotor blade. As shown in FIG. **28**, boom **510** may include a reinforcement web **514** to improve lateral stability of the boom. Reinforcement web **514** may extend from the boom mount end of boom **510** to the rotor end of boom **510** and may extend along the width of boom **510** along the illustrated 'Y' axis. As shown, reinforcement web **514** can connect respective load-bearing inserts **1660** of a boom to provide increased lateral stability of each boom **510**. Boom **510** embodiments constructed as composite laminates may have load-bearing inserts **1660** which are co-molded during lamination or mounted as a post-cure secondary operation.

(161) FIG. **29** depicts a perspective view of boom mount **1630**. In contrast with boom mount **163**, boom mount **1630** is designed so that the mounting end of boom **510** slides into a respective opening in the boom mount structure, rather than over the boom mount structure as with boom mount **163**. Boom mount **1630** also differs from boom mount **163** in that it consists of both an external element (exoskeleton shell **1632**) and an internal element (endoskeleton assembly **1634**). In the exemplary embodiment shown in FIG. **29** and FIG. **30**, the exoskeleton shell **1632** is a reinforced composite laminate monocoque and the endoskeleton assembly **1634** is a collection of machined metal components, but other exoskeleton and endoskeleton materials and mechanical solutions are possible. As shown in FIG. **30**, when a boom **510** is mounted to a respective opening in the boom mount **1630**, fasteners pass thru the load-bearing inserts **1660** in the boom **510** mounting end and into the endoskeleton lateral members **1638**, providing an exemplary mechanical joint and a rigid powerplane structure.

(162) Notably, endoskeleton lateral members **1638** directly connect the side fasteners of opposing booms with the shortest line of tension, while the exoskeleton shell **1632** directly connects opposing top and bottom boom fasteners, effectively joining two opposing booms into a continuous, single beam element. Similarly, the short, direct line of tension in the exoskeleton shell **1632** between adjacent boom **510** side fasteners and load-bearing inserts **1660**, in combination with the stabilizing effect of endoskeleton lateral members **1638**, more effectively restrains lateral movement of the cantilevered booms than boom mount **163**. Other boom mount **1630** structural solutions which don't use an exoskeleton or endoskeleton are possible, including tubular spaceframe weldments and 3D printed generative design frameworks. For example, in some embodiments, boom mount **1630** can be implemented as a 3D printed frame in which the mechanical functions of the exoskeleton shell **1632** and endoskeleton assembly **1634** are combined in a unitary structure.

(163) FIGS. **29** and **30** show an exemplary pivotal coupling **1520** mounted to the bottom of boom mount **1630**. Functionally similar to pivotal coupling **152**, pivotal coupling **1520** extends from the bottom of boom mount **1630** and connects the powerplane assembly **330** with fuselage **250** of aircraft **200** by means of a biaxial pivot that permits the powerplane assembly **330** to pivot about the pitch and roll axes of aircraft **200**. In appearance, pivotal coupling **1520** is broadly similar to a common driveshaft universal joint assembly, and similarly consists of two yokes joined by a cross, but pivotal coupling **1520** is mechanically optimized for the unique requirements of use in aircraft **200** including tensile strength and angular misalignment capability, rather than optimizing for torque transmission as is common with a driveshaft universal joint. Compared with pivotal coupling **152**, pivotal coupling **1520** is more compact, has a more direct load path with less deflection, and is easier to manufacture with less touch labor. FIG. **31** depicts a perspective view of pivotal coupling **1520**, showing the upper yoke **1730** and its backing plate **1550A**, the one-piece bearing cross **1770**, and the lower yoke **1710** and its backing plate **1550B**. As shown in FIG. **30**, the upper yoke flange **1750A** and its backing plate **1550A** sandwich the bottom of the boom mount

exoskeleton shell **1632**, secured with a ring of fasteners through apertures **1690A** to form a rigid coupling. Further, four of the upper yoke flange **1750A** mounting fasteners connect the yoke directly to endoskeleton lateral members **1638** by means of endoskeleton vertical members **1636**, helping to transfer the fuselage **250** load to the booms **510** while also helping to vertically stiffen boom mount **1630**. Likewise, the lower yoke flange **1750B** and its backing plate **1550B** sandwich the upper bulkhead of the powerpack assembly **320**, again secured by a ring of fasteners through apertures **1690B** to form a rigid coupling. Other mounting solutions appropriate for other airframe and boom mount structures such as tubular spaceframes are possible.

(164) By this means, pivotal coupling **1520** shown in FIG. **31** functions as a biaxial connector that permits the powerplane assembly **330** to partially rotate (pivot) separately or simultaneously around the axes of pivot pin **1810** and pivot pin **1830**, thereby enabling the powerplane assembly **330** to pitch and roll, separately or simultaneously, with respect to aircraft fuselage **250** as required for vehicle navigation and control, while the powerplane and aircraft fuselage remain locked together around the yaw axis and while the aircraft fuselage **250** remains in its primary horizontal or vertical orientation (depending on vehicle configuration). In other words, as previously described, differential rotor thrusting may be used to pitch the powerplane assembly **330** forward (nose down) or backward (nose up), or to roll the powerplane assembly **330** left or right, and to make these pitch and roll movements either independently or simultaneously with each other in any combination, all without the use of mechanical actuators (pneumatic, hydraulic or electric), and all while the pivoting powerplane **330** remains indexed relative to the fuselage **250** along the yaw axis and while the aircraft fuselage **250** remains in its primary horizontal or vertical orientation. Further, differential rotor torque inputs can be used to rotate the entire aircraft **200** about its yaw axis as desired.

(165) In other embodiments, other types of biaxial or non-biaxial couplings may be used to pivotally connect the powerplane assembly **330** to the fuselage **250**, including a wide range of mechanical couplings such as spherical bearings and hinges, or compliant, flexible couplings such as wire rope, webbing or reinforced rubber. Yet other types of couplings are possible in other embodiments to connect the powerplane assembly **330** to the fuselage **250** while permitting relative movement of the powerplane assembly **330** with respect to the fuselage **250** about one or more axes (e.g., the pitch and roll axes).

(166) During operation, the flight controller **63** (FIG. **24**) is configured to receive input from the sensors **69** (FIG. **24**) and to control flight of the aircraft **200** based on the sensor inputs. As an example, the sensors **69** may indicate aircraft attitude and movement (e.g., airspeed), and the flight controller **63** may provide control signals to the motor controllers **65** for controlling the rotors **520** to generate the appropriate thrust for achieving a desired attitude of the powerplane assembly **330** and movement of the aircraft **200**. At least some sensors **69** (e.g., a GPS sensor) may be used to sense the location of the aircraft **200**, and the flight controller **63** may control movement of the aircraft **200** in order to navigate the aircraft **200** to a desired destination. At least some sensors **69**, such as radar or LIDAR, may be used to sense objects external to the aircraft **200**, and the flight controller **63** may use information from these types of sensors to sense and avoid collision threats.

(167) Based on the input from the sensors **69**, the thrusts generated by the rotors **520** are controlled to move the aircraft **200** as desired to a destination. As an example, thrust may be increased to lift the aircraft **200** off the ground. After taking off, differential thrusting may be used to control the direction of flight and attitude of the powerplane assembly **330**. Thrusting may also be used to control the ascent and descent of the aircraft **200** as may be desired. As the attitude of the powerplane assembly **330** is changing about the pitch axis and roll axis, the fuselage **250** hangs from the pivotal coupling **1520** and maintains a vertical orientation due to gravity, independent of the movements of the powerplane assembly **330**.

(168) Once the aircraft **200** reaches its destination, the flight controller **63** controls the rotors **520** so that the aircraft **200** gently lands on the docking station **900** or, in some embodiments, hovers just

above ground level (FIG. 34). If an emergency condition is detected during the flight, such as a detection of a fire or failure of a critical component (e.g., one or more motors **660**), the flight controller **63** may be configured to actuate the BRS **72** in order to deploy a parachute that allows the aircraft **200** to slowly and safely return to the ground. In other embodiments, other techniques and/or recovery systems for handling an emergency condition are possible, such as the flight controller **63** activating a secondary motor system to provide functionality in place of a failed motor **660**.

Example Uses

(169) As described herein, electrically powered vertical takeoff and landing (VTOL) aircraft feature the ability to take-off and land without a runway, and may have other capabilities well suited for operations in traditionally restricted urban areas.

(170) Take-off and landing activities are an important component of safe VTOL operation in urban areas, and involve a variety of logistical considerations such as pedestrian and passenger safeguarding, obstacle-free airspace, vehicle maneuvering, vehicle noise, vehicle downwash, the local micro-climate and wind environment, etc. as well as commercial considerations such as site location, customer access, service hours, service site distribution, ground support crew requirements, business scalability, etc.

(171) Further, many aspects of take-off and landing are coupled to vehicle design and performance. For example, high take-off mass vehicles often generate more noise and downwash than lighter vehicles, and winged vehicles may be less maneuverable than pure rotorcraft or may require a dedicated airspace zone to safely transition between vertical and horizontal flight.

(172) The following sections describe an example embodiment VTOL aircraft **200** and a dedicated VTOL docking station **900** designed to receive aircraft **200**. Together, the synergistic aircraft and VTOL docking station may be used to provide a novel 3D mobility system useful for transporting passengers or cargo in urban or suburban airspace traditionally subject to aircraft restrictions due to noise, safety or vehicle maneuverability concerns. Some embodiments of the present invention are illustrated as an example and are not limited by the figures of the accompanying drawings.

(173) FIG. 32 is a simplified, system-level illustration of a novel 3D mobility system method of operation **S100** consisting of an example embodiment aircraft **200** and an example embodiment VTOL docking station **900** designed to work together as a synergistic system.

(174) Step **S110** shows a representative aircraft **200** approaching a representative VTOL docking station **900**. The VTOL docking station **900** is self-enclosed and fully autonomous, with a controlled-axis automatic door that provides passenger access to the aircraft (e.g., VTOL aircraft **200**) during boarding or de-boarding operations, thereby ensuring a safe, secure, unobstructed landing zone for representative aircraft **200**. Similarly, nearby pedestrians are shielded and protected from the aircraft during take-off and landing operations by the enclosed VTOL docking station **900** with closed automatic door. The representative VTOL docking station **900** may be readily installed in a wide variety of locations with little site preparation and does not require a human operator. The representative VTOL docking station **900** also provides a fixed, defined landing location for autonomous aircraft **200**, thereby facilitating the use of defined, pre-programmed, precision flight paths from one station to the next and reducing the need for more sophisticated navigational and sense and avoid technologies.

(175) Step **S120** shows representative aircraft **200** hanging at rest in the VTOL docking station **900**, with the automatic doors on both the aircraft and VTOL docking station open for boarding. Note that representative aircraft **200** has no landing gear. Instead, the aircraft hangs from the structural booms **510** of the powerplane assembly **330**, which rest on the load-bearing rim of the VTOL docking station **900**. If the aircraft requires electrical charging after docking, the VTOL docking station door closes behind the previous departing passenger and remains closed until the charging operation is complete and the vehicle is ready for service.

(176) Precision landing in the example VTOL docking station is made possible by the unique

pivoting powerplane **330** technology of representative aircraft **200**, which can quickly tilt in a wide range of angles (e.g., ± 30 degrees) in any compass direction, enabling the aircraft to hold station above a designated grid reference such as VTOL docking station **900** even in gusty or turbulent wind conditions. As used herein “gusty, changing wind conditions” means winds of up to approximately 20 to 30 mph. Further, because the pivoting powerplane **330** is located at the upper extremity of the vertically elongated fuselage of aircraft **200**, the powerplane may continue to tilt and adjust position as necessary while the fuselage **250** descends into the VTOL docking station, ensuring that the vehicle does not collide with the docking station walls and that the structural booms come to rest smoothly on the VTOL docking station rim. Also note that VTOL docking station **900** and the unique vertically elongated fuselage of representative aircraft **200** are designed to place the vehicle floor at approximately the same elevation as the ground when the vehicle is docked, enabling the passenger to effortlessly step directly into or out of the vehicle, similar to entering or exiting an elevator.

(177) Step **S130** shows the representative aircraft **200** in-flight along a pre-determined flight path between two representative VTOL docking stations **900**. Note that both docking station doors are closed to prevent tampering and to secure the takeoff/landing zones, and that the unique pivoting powerplane technology of representative aircraft **200** enables it to fly an energy optimal 3D flight path with a constant heading, irrespective of changing wind conditions.

(178) FIG. **33** shows an exemplary method of operation **S200** using the same example aircraft **200** and VTOL docking station **900** as discussed with respect to FIG. **32**, but with the destination VTOL docking station **900** installed inside a building. In FIG. **33**, the upper portion of the example VTOL docking station **900** protrudes through an opening in the roof of the building and passengers can advantageously access the vehicle directly from inside the building while the vehicle noise and downwash remain outside. Suitable locations for exemplary indoor VTOL docking stations include airports, hotels, transit terminals, shopping malls, office and residence complexes, and many other possibilities. As shown in FIG. **33**, in step **S210**, a passenger is able to board aircraft **200** which is docked with VTOL docking station **900**. Step **S220** shows the aircraft **200** in flight between respective VTOL docking stations **900**. Step **S230** shows a passenger upon disembarking from the destination VTOL docking station **900** that is located inside the building.

(179) FIG. **34** depicts another exemplary method of use of VTOL docking station **900** and an exemplary aircraft **200**, with aircraft **200** featuring advanced navigation and sense and avoid technology. In this enhanced exemplary embodiment of a point-to-point, on-demand 3D transportation system, in step **S310**, an empty aircraft **200** is assigned the customer's location by a smartphone app and takes off from a VTOL docking station **900** in which it was docked. In step **S320**, the aircraft **200** may use a hybrid multi-modal navigation system and advanced sense and avoid capabilities (e.g., utilizing onboard sensors **69**) to autonomously navigate to the customer's location. The low-inertia powerplane **330** pivots responsively to enable the aircraft **200** to hold the specified geospatial grid reference location even in variable winds. The elevated powerplane **320** minimizes downwash interaction as the aircraft **200** hovers over the designated target and begins its descent. The vehicle may utilize advanced sense and avoid technology to safely descend to a pre-determined hover altitude where the floor of the vehicle is at a specified distance above the ground. Holding station in near-ground hover, the aircraft **200** automatic door **410** opens, the customer steps up into aircraft **200**, the automatic door **410** closes, and aircraft **200** ascends and navigates autonomously to a destination docking station **900** in step **S330**.

(180) Notably, the enhanced, exemplary point-to-point solution described with respect to FIG. **34** is made possible by the core technologies described with respect to aircraft **200**, including: (1) a low-inertia, passively pivoting powerplane (e.g., powerplane assembly **330**) that enables aircraft **200** to hold 3D station above a defined grid reference even in variable winds; (2) an elevated powerplane (e.g., powerplane assembly **330**) that minimizes ground-effect induced perturbations and enhances near-ground stability; (3) an elevated powerplane (powerplane assembly **330**) that enables a

passenger to safely enter and exit the vehicle while the rotors are turning; (4) a stable fuselage **250**/cabin **310** that is isolated from powerplane movement by a biaxial pivotal coupling (e.g., pivotal coupling **1520**); (5) an automatic door (e.g., door **410**); and (6) a low floor height easily reachable during near-ground aircraft hover.

(181) In summary, a novel 3D mobility system that provides customers with safe, low cost, stable, quiet, widely distributed and easily accessible 3D transportation, with a focus on relatively short, low-altitude journeys in urban or suburban environments, is disclosed herein. The embodiments disclosed herein provide for a VTOL service in which VTOL aircraft are capable of autonomous flight between docking stations and autonomous charging upon docking.

(182) As previously noted above, the exemplary aircraft designs described herein have many advantages relative to conventional aircraft. As an example, the exemplary aircraft described herein have a relatively simple and efficient design helping to reduce manufacturing and operating costs, yet they have many safety features that prevent injury and allow reliable and effective recovery from various emergency events. Indeed, the aircraft are capable of withstanding various types of failures and protecting a passenger in the event of an electrical fire. In addition, the powerplane assembly **33** and powerplane assembly **330** are located high off the ground such that inadvertent contact between the rotors **52** or rotors **520** and people on the ground is prevented. The aircraft described herein can operate using clean energy with relatively low noise and provide VTOL capability thereby permitting use of the aircraft in urban areas where noise and access to suitable landing sites are concerns. The aircraft also has a relatively small horizontal footprint, making it suitable for use in areas with limited space, particularly in an urban environment. Various other benefits and advantages are described herein and would also be apparent to a person of ordinary skill upon reading this disclosure.

(183) The foregoing is merely illustrative of the principles of this disclosure and various modifications may be made by those skilled in the art without departing from the scope of this disclosure. The above described embodiments are presented for purposes of illustration and not of limitation. The present disclosure also can take many forms other than those explicitly described herein. Accordingly, it is emphasized that this disclosure is not limited to the explicitly disclosed methods, systems, and apparatuses, but is intended to include variations to and modifications thereof, which are within the spirit of the following claims.

(184) As a further example, variations of apparatus or process parameters (e.g., dimensions, configurations, components, process step order, etc.) may be made to further optimize the provided structures, devices, and methods, as shown and described herein. In any event, the structures and devices, as well as the associated methods, described herein have many applications. Therefore, the disclosed subject matter should not be limited to any single embodiment described herein, but rather should be construed in breadth and scope in accordance with the appended claims.

(185) The terms “about” and “approximately” shall generally mean an acceptable degree of error or variation for the quantity measured given the nature or precision of the measurements. Typical, exemplary degrees of error or variation are within 20 percent (%), preferably within 10%, more preferably within 5%, and still more preferably within 1% of a given value or range of values. Numerical quantities given in this description are approximate unless stated otherwise, meaning that the term “about” or “approximately” can be inferred when not expressly stated.

(186) With reference to the use of the word(s) “comprise,” “comprises,” and “comprising” in the foregoing description and/or in the following claims, unless the context requires otherwise, those words are used on the basis and clear understanding that they are to be interpreted inclusively, rather than exclusively, and that each of those words is to be so interpreted in construing the foregoing description and/or the following claims.

(187) The term “including” should be interpreted to mean “including but not limited to . . .” unless the context clearly indicate otherwise.

(188) The term “consisting essentially of” means that, in addition to the recited elements, what is

claimed may also contain other elements (steps, structures, ingredients, components, etc.) that do not adversely affect the operability of what is claimed for its intended purpose. Such addition of other elements that do not adversely affect the operability of what is claimed for its intended purpose would not constitute a material change in the basic and novel characteristics of what is claimed.

(189) The term “adapted to” means designed or configured to accomplish the specified objective, not simply able to be made to accomplish the specified objective.

(190) The term “capable of” means able to be made to accomplish the specified objective.

(191) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well (i.e. “at least one”), unless the context clearly indicates otherwise.

(192) The terms “first”, “second”, and the like are used herein to describe various features or elements, but these features or elements should not be limited by these terms. These terms are only used to distinguish one feature or element from another feature or element. Thus, a first feature or element discussed below could be termed a second feature or element, and similarly, a second feature or element discussed below could be termed a first feature or element without departing from the teachings of the present disclosure.

(193) Terms such as “at least one of A and B” should be understood to mean “only A, only B, or both A and B.” The same construction should be applied to longer list (e.g., “at least one of A, B, and C”).

Claims

1. A vertical takeoff and landing (VTOL) aircraft system, comprising: a VTOL aircraft comprising: a pivotal coupling comprising a boom mount; a fuselage coupled to the boom mount, the fuselage comprising a cabin; a powerplane assembly comprising: a plurality of rotors each coupled to an electrical motor; a plurality of booms directly coupled to the boom mount and coupling the plurality of rotors to the boom mount, each of the plurality of booms comprising a boom contact strip located on a bottom surface of a respective boom; at least one aircraft battery pack electrically coupled to the electrical motors of the plurality of rotors and electrically coupled to each boom contact strip of the plurality of booms; a VTOL docking station comprising: one or more docking station contact strips located on an upper surface of the VTOL docking station and configured to make electrical contact with a respective boom contact strip of a plurality of boom contact strips.
2. The VTOL aircraft system of claim 1, wherein the VTOL docking station further comprises: electrical charging equipment electrically coupled to the one or more docking station contract strips, wherein the VTOL docking station is further configured to charge the at least one aircraft battery pack when the one or more docking station contact strips make electrical contact with the plurality of boom contact strips.
3. The VTOL aircraft system of claim 2, wherein: the VTOL aircraft further comprises an aircraft charge controller electrically coupled to the at least one aircraft battery pack; the VTOL docking station further comprises a docking station controller in communication with the electrical charging equipment, wherein the docking station controller is configured to communicate with the aircraft charge controller to control charging of the at least one aircraft battery pack.
4. The VTOL aircraft system of claim 1, wherein: each boom of the plurality of booms comprises a boom mount end comprising an upper fastener, a lower fastener, and two side fasteners; each boom of the plurality of booms has an opposing boom; and the boom mount comprises: an endoskeleton directly connecting respective side fasteners of opposing booms; and an exoskeleton directly connecting respective upper fasteners and respective lower fasteners of opposing booms and directly connecting respective side fasteners of adjacent booms.

5. The VTOL aircraft system of claim 1, wherein each boom of the plurality of booms comprises a boom mount end comprising an approximately diamond profile and a rotor end, wherein each boom mount comprises a first taper between the boom mount end and the rotor end and a second taper between the boom mount end and the rotor end.
6. The VTOL aircraft system of claim 5, wherein the first taper comprises a vertical taper and the second taper comprises a horizontal taper.
7. The VTOL aircraft system of claim 5, wherein each boom is hollow and includes a reinforcement member connecting the boom mount end and the rotor end and extending across a width of the boom.
8. The VTOL aircraft system of claim 1, wherein: the pivotal coupling permits the powerplane assembly to passively pivot relative to the cabin about a pitch axis and a roll axis; the cabin is indexed to the powerplane assembly about a yaw axis; and the yaw axis is perpendicular to the pitch axis and the roll axis.
9. The VTOL aircraft system of claim 1, wherein the VTOL aircraft further comprises: a plurality of motor controllers coupled to the electrical motors of the plurality of rotors; and a flight controller configured to communicate with the plurality of motor controllers for controlling rotor speed of the plurality of rotors.
10. The VTOL aircraft system of claim 1, wherein the VTOL docking station further comprises a docking station frame configured to support the VTOL aircraft.
11. The VTOL aircraft system of claim 10, wherein the docking station frame comprises a material selected from aluminum, steel, carbon fiber, and combinations thereof.
12. The VTOL aircraft system of claim 1, wherein the VTOL aircraft is configured to dock with the VTOL docking station such that the plurality of booms make contact with the upper surface of the VTOL docking station and the cabin is contained within an interior of the VTOL docking station.
13. The VTOL aircraft system of claim 1, wherein the VTOL docking station comprises a cylindrical shape configured to contain the cabin within an interior of the VTOL docking station when the VTOL aircraft is docked with the VTOL docking station.
14. The VTOL aircraft system of claim 1, further comprising: a barrier separating an interior of the VTOL docking station from an environment external to the VTOL docking station; a door configured to provide access to and from the interior of the VTOL docking station; and an access control mechanism that is configured to transition the door between a locked state and an unlocked state and to cause the door to transition between an open configuration and a closed configuration.
15. The VTOL aircraft system of claim 14, wherein the access control mechanism is configured to transition the door to the unlocked state in response to the VTOL aircraft docking with the VTOL docking station and subsequently reducing generated thrust beyond a predetermined threshold.
16. The VTOL aircraft system of claim 14, wherein the access control mechanism is configured to transition the door to the locked state in response to the VTOL aircraft initiating takeoff from the VTOL docking station.
17. The VTOL aircraft system of claim 14, wherein the access control mechanism is configured to transition the door to the locked state in response to the VTOL aircraft increasing generated thrust beyond a predetermined threshold.
18. The VTOL aircraft system of claim 14, wherein the access control mechanism is configured to wirelessly receive lock and unlock commands from a user device associated with a passenger of the VTOL aircraft.
19. The VTOL aircraft system of claim 1, wherein the at least one aircraft battery pack is positioned between the cabin and the powerplane assembly.
20. The VTOL aircraft system of claim 1, wherein the VTOL aircraft comprises a center of mass positioned between the cabin and the powerplane assembly.
21. The VTOL aircraft system of claim 1, wherein the fuselage is configured to maintain an approximately vertical orientation during lateral flight.

22. A vertical takeoff and landing (VTOL) aircraft, comprising: a pivotal coupling comprising a boom mount; a fuselage coupled to the boom mount, the fuselage comprising a cabin; a powerplane assembly comprising: a plurality of rotors each coupled to an electrical motor, each of the plurality of rotors including a boom mount end and a rotor end, wherein each of the plurality of rotors includes a first taper between the boom mount end and the rotor end and a second taper between the boom mount end and the rotor end; a plurality of booms directly coupled to the boom mount and coupling the plurality of rotors to the boom mount; a plurality of motor controllers coupled to the electrical motors of the plurality of rotors; a flight controller configured to communicate with the plurality of motor controllers for controlling rotor speed of the plurality of rotors; wherein the pivotal coupling permits the powerplane assembly to passively pivot relative to the cabin about a pitch axis and a roll axis; wherein the cabin is indexed to the powerplane assembly about a yaw axis; and wherein the yaw axis is perpendicular to the pitch axis and the roll axis.

23. The VTOL aircraft of claim 22, wherein the VTOL aircraft comprises at least one aircraft battery pack within the fuselage, the at least one aircraft battery pack electrically coupled to the electrical motors of the plurality of rotors and positioned above the cabin between the cabin and the pivotal coupling.

24. The VTOL aircraft of claim 22, wherein each boom of the plurality of booms comprise a boom contact strip located on a bottom surface of a respective boom.

25. The VTOL aircraft of claim 22, wherein: the boom mount end of each boom further comprises an upper fastener, a lower fastener, and two side fasteners; each boom of the plurality of booms has an opposing boom; and the boom mount comprises: an endoskeleton directly connecting respective side fasteners of opposing booms; and an exoskeleton directly connecting respective upper fasteners and respective lower fasteners of opposing booms and directly connecting respective side fasteners of adjacent booms.

26. The VTOL aircraft of claim 22, wherein the first taper comprises a vertical taper and the second taper comprises a horizontal taper.

27. The VTOL aircraft of claim 22, wherein each boom is hollow and includes a reinforcement member connecting the boom mount end and the rotor end.

28. The VTOL aircraft of claim 22, wherein the at least one aircraft battery pack is positioned between the cabin and the powerplane assembly.

29. The VTOL aircraft of claim 22, wherein the VTOL aircraft comprises a center of mass positioned between the cabin and the powerplane assembly.

30. The VTOL aircraft of claim 22, wherein the fuselage is configured to maintain an approximately vertical orientation during lateral flight.

31. The VTOL aircraft of claim 22, wherein the fuselage is configured to maintain an approximately vertical orientation during high gust wind conditions.

32. A method, comprising: generating thrusts with a powerplane assembly comprising plurality of rotors mounted on a vertical takeoff and landing (VTOL) aircraft, each of the plurality of rotors comprising at least one rotor blade and an electrical motor of a plurality of electrical motors for driving the at least one rotor blade, wherein each of the plurality of rotors is coupled to a boom mount of a pivotal coupling via a boom extending from the boom mount, wherein each boom is directly coupled to the boom mount and comprises a double taper between a boom mount end and a rotor end of the boom; and controlling the plurality of electrical motors to generate differential thrust such that the powerplane assembly (1) pivots about a pitch axis and a roll axis relative to a fuselage of the VTOL aircraft having a cabin and (2) is indexed to the cabin about a yaw axis, wherein the yaw axis is perpendicular to the pitch axis and the roll axis.

33. The method of claim 32, wherein: each boom of the plurality of booms comprises a boom contact strip located on a bottom surface of a respective boom; and the VTOL aircraft further comprises at least one aircraft battery pack electrically coupled to the electrical motors of the plurality of rotors and electrically coupled to each boom contact strip of the plurality of booms.

34. The method of claim 33, further comprising: docking the VTOL aircraft with a VTOL docking station thereby causing the contact strip of each boom to make electrical contact with one or more docking station contact strips of the VTOL docking station; and causing an electrical charging component of the VTOL docking station to charge the at least one aircraft battery pack via the electrical contact between the one or more docking station contact strips and each boom contact strip.

35. The method of claim 34, wherein docking the VTOL aircraft with the VTOL docking station further comprises suspending the VTOL aircraft on the VTOL docking station by making contact between the plurality of booms and an upper surface of the VTOL docking station.

36. The method of claim 34, wherein the fuselage is configured to maintain an approximately vertical orientation during takeoff, landing, and flight.

37. The method of claim 36, wherein the VTOL aircraft is further configured to maintain the approximately vertical orientation during high gust wind conditions.

38. The method of claim 34, further comprising: the VTOL aircraft taking off from the VTOL docking station; and the VTOL aircraft autonomously flying between the VTOL docking station to a second VTOL docking station.

39. The method of claim 34, further comprising: the VTOL aircraft receiving a position associated with a passenger and a target destination; the VTOL aircraft taking off from the VTOL docking station; the VTOL aircraft autonomously flying to the position associated with the passenger; the VTOL aircraft hovering in place above a ground surface while the passenger boards the VTOL aircraft; and the VTOL aircraft autonomously flying to the target destination.

40. The method of claim 39, wherein the target destination is associated with a destination VTOL docking station.

41. The method of claim 39, wherein the target destination does not include a destination VTOL docking station and the VTOL aircraft autonomously flying to the target destination further comprises the VTOL aircraft hovering in place above a ground surface of the target destination while the passenger deboards the VTOL aircraft.

42. The method of claim 34, further comprising: the VTOL aircraft receiving, from a passenger, a target destination associated with a destination VTOL docking station; the passenger boarding the VTOL aircraft while docked at the VTOL docking station; and the VTOL aircraft autonomously flying to the destination VTOL docking station.

43. The method of claim 32, wherein: the boom mount end of each boom further comprises an upper fastener, a lower fastener, and two side fasteners; each boom of the plurality of booms has an opposing boom; and the boom mount comprises: an endoskeleton directly connecting respective side fasteners of opposing booms; and an exoskeleton directly connecting respective upper fasteners and respective lower fasteners of opposing booms and directly connecting respective side fasteners of adjacent booms.

44. A vertical takeoff and landing (VTOL) docking station comprising: a barrier configured to support a VTOL aircraft; one or more linear or curvilinear docking station contact strips located on an upper surface of the barrier wherein the one or more linear or curvilinear docking station contact strips make electrical contact with complementary contact strips of the VTOL aircraft.

45. The VTOL docking station of claim 44, further comprising: electrical charging equipment electrically coupled to the one or more docking station contract strips, wherein the VTOL docking station is further configured to charge at least one aircraft battery pack of the VTOL aircraft when the one or more docking station contact strips make electrical contact with complementary contact strips of the VTOL aircraft.

46. The VTOL docking station of claim 45, further comprising a docking station controller in communication with the electrical charging equipment, wherein the docking station controller is configured to communicate with an aircraft charge controller of the VTOL aircraft to control charging of the at least one aircraft battery pack.

47. The VTOL docking station of claim 44, wherein each of the one or more docking station contact strips comprise an arc angle of about 60 degrees.
48. The VTOL docking station of claim 44, wherein the VTOL docking station is configured to contain a cabin of the VTOL aircraft when the VTOL aircraft is docked with the VTOL docking station.
49. The VTOL docking station of claim 48, wherein the VTOL docking station comprises a cylindrical shape configured to contain the cabin when the VTOL aircraft is docked with the VTOL docking station.
50. The VTOL docking station of claim 44, wherein the VTOL docking station comprises a docking station frame configured to support the VTOL aircraft.
51. The VTOL docking station of claim 44, further comprising: a barrier separating an interior of the VTOL docking station from an environment external to the VTOL docking station; a door configured to provide access to and from the interior of the VTOL docking station; and an access control mechanism that is configured to transition the door between a locked state and an unlocked state and to cause the door to transition between an open configuration and a closed configuration.
52. The VTOL docking station of claim 51, wherein the access control mechanism is configured to transition the door to the unlocked state in response to the VTOL aircraft docking with the VTOL docking station and subsequently reducing generated thrust beyond a predetermined threshold.
53. The VTOL docking station of claim 51, wherein the access control mechanism is configured to transition the door to the locked state in response to the VTOL aircraft initiating takeoff from the VTOL docking station.
54. The VTOL docking station of claim 51, wherein the access control mechanism is configured to transition the door to the locked state in response to the VTOL aircraft increasing generated thrust beyond a predetermined threshold.
55. The VTOL docking station of claim 51, wherein the access control mechanism is configured to wirelessly receive lock and unlock commands from a user device associated with a passenger of the VTOL aircraft.
56. The VTOL docking station of claim 44, wherein the VTOL docking station is configured to be installed indoors.
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